

Circular Capital in the Artificial Intelligence Stack: A Game-Theoretic Analysis of Market Power, Financial Fragility, and Welfare

Abstract

The 2024–2026 artificial-intelligence investment boom exhibits the financing structure that preceded previous round-trip busts: dense, committed circular capital in which upstream investment returns downstream as contracted compute spend, measured here at \$675.40 billion across 17 committed contracts. This architecture concentrates burst incidence upstream. While bubble indicators peak at $\sim 83\%$ among frontier labs where valuations have detached from revenue, credit markets already price stack leverage through a record single-name CDS spread (215 bp) and downgrade. Six solved bilateral games reveal an aggregate deadweight loss of \$416.77 billion (71.28% efficiency), with non-cooperation binding in the Hardware–Cloud bottleneck. Consequently, a severe repricing destroys $\sim \$239$ billion of Hardware and $\sim \$133$ billion of Cloud value first, transmitting a $\sim \$139$ billion loss ($\sim 0.45\%$ of GDP) to the U.S. economy via capex and wealth channels. This loss hierarchy is structural: it survives 16 parameter extremes and 300 joint Monte Carlo draws, and reproduces the historical loss order when applied blind to 2006–07 pre-crisis inputs from the 2008 Global Financial Crisis. By benefit–cost ratio, optimal policy sequencing prioritizes contract disclosure and lending caps. Three empirical triggers would cancel the forecast if observed.

Keywords: artificial intelligence; financial bubbles; circular dealing; vendor financing; game theory; burst prediction; financial stability; competition policy.

JEL codes: D43, G01, G12, G14, L13, L41, E44, O33.

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1 Introduction

This paper converts a measured financing structure into a dated, testable prediction: an AI-asset repricing is likely while the current committed spend cycle runs; its losses will land first on Hardware, then Cloud, then frontier labs, then apps; and the severe case costs the US economy roughly \$100–150 billion. Section 2 states that forecast as eight numbered claims, each with an assessed likelihood (or an explicit abstention, P8) and a pointer to the observable that would overturn it. The remaining sections supply the evidence (Sections 5–11), the record of stress testing and historical validation (Sections 12–14), and the investment and policy implications (Sections 15–19). An Online Appendix documents the grading apparatus, assumptions, and caveats; the body below advances the predictions without interruption.

The argument is a single chain with five links. **Measurement:** \$675.40 billion of committed circular capital is read off signed contracts, not inferred from sentiment or multiples (Section 5). **Mechanism:** six solved bilateral games show non-cooperation binds in exactly one cell — Hardware versus Cloud — trapping value upstream (Sections 6–7). **Incidence:** the trap fixes the burst order — Hardware (the Toolmakers), Cloud (the Landlords), frontier labs (the Brain Builders), apps (the App Builders) — with formation conditions worst where private marks have detached furthest (Sections 9–10). **Validation:** the order survives sixteen extremes and 300 joint draws; the same modules recover the 2008 loss order, and on telecom equipment in 2001 they recover the mechanism — the stuck vendor cell, the first-to-fail ranking, the formation order — while missing dollar incidence, which multiple compression decides outside contract arithmetic (Sections 12–14). **Action:** triggers monitor the preconditions and remedies sequence by benefit–cost ratio, disclosure and lending caps first (Sections 15–19). Break any link and the conclusion falls; Section 20 names the observables that would break each one. One distinction governs every number that follows: *rankings* are the high-confidence claims; *dollar figures* mark scenario scale for sizing remedies.

2 Central Forecast: Repricing Likelihood, Loss Incidence, and Macroeconomic Cost

This paper makes a prediction with three parts. **First**, an AI-asset repricing is likely while the current committed spend cycle runs: the conditions that have preceded every modern round-trip bust — dense reciprocal financing, private marks detached from capturable revenue, and credit markets starting to charge for leverage — are all present at once. **Second**, the damage has a fixed address: Hardware first (~\$239B), Cloud second (~\$133B), frontier labs third (~\$85B), app companies last (~\$21B), because contracts, not sentiment, decide where losses land. **Third**, the US GDP cost of the severe case is ~\$139B (~0.45%), composed of canceled tech investment (~\$120B) plus the wealth effect of falling equities (~\$19B).

What this paper does *not* predict is a date. Timing belongs to funding rounds, credit events, and capex guidance — the triggers of Section 15 — not to models. Everything else here is a measured claim, not a scenario sketch: the capital structure is read off contracts, the binding constraint is proved in solved games, and the resulting incidence ranking is the same order that survives every stress battery and transfers to 2008. The result is a standing warning with an address book: it tells you what breaks, in what order, at roughly what scale, and what to do before it happens.

How seriously to take each claim. Every numbered prediction below carries two separate tags. **Likelihood** is the author’s assessed probability that the claim proves right, on the fixed scale in Table 1 — a judgment in the forecasting tradition (IPCC, Tetlock), not a model esti-

mate. **Confidence** rates the quality of the evidence behind that judgment (High = contracts plus surviving stress tests; Medium = scenario arithmetic that moves with inputs; Low = no evidentiary basis). Model vote-shares (16/16, 100%, 65%) measure *robustness*, and formation scores (82.9% etc.) measure *vulnerability* — neither is the probability of the event. The two numbers that look most like probabilities in this paper are therefore the two you must not mistake for forecasts; the actual event probabilities are the author’s, stated in the registry.

Table 1. Likelihood scale for author-assessed probabilities. Used only in the registry below; nowhere else in the paper does a percent sign denote an event probability.

Wording	Assessed probability
Very likely	85–95%
Likely	65–80%
More likely than not	50–65%
Toss-up	~50%
Unlikely	below 35%

Table 2. Prediction registry: every claim, its likelihood, its evidence grade, and what retires it.

ID	Claim	Likelihood	Confidence	Overtaken by
P1	AI-asset repricing before the committed spend cycle exhausts itself	Likely (~65%)	Medium	Validation tests 1–2
P2	<i>Conditional on a severe burst</i> , losses rank Hardware > Cloud > labs > apps	Very likely (~90%)	High	Test 3 (ranking flips)
P3	Severe GDP cost lands near \$100–150B (~0.3–0.5%), mostly via canceled capex	More likely than not (~55%)	Medium	A burst with no capex cut
P4	Frontier-lab funding stress is the nearest trigger	Likely (~70%)	Medium	A first break elsewhere (Sec. 15)
P5	Labs carry the highest bubble vulnerability (formation rank FM first)	More likely than not (~60%)	Medium	Formation rank flips (65% joint agreement)
P6	Round two is small (~\$15B), mostly on labs (~\$13.9B of ~\$15.2B)	More likely than not (~60%)	Medium	Collateral fire sale (Sec. 11)
P7	Disclosure first, lending caps second is the optimal remedy sequence (caps still rank first by BCR)	Likely (~70%)	Medium	BCR rank reversal (Sec. 19)
P8	Any calendar date for the repricing	— (unassigned)	Low	— (no date predicted)

P8 stays unassigned by design: structural game theory identifies equilibrium fragility and loss topology, while the tipping date depends on non-deterministic catalytic triggers (funding-round failures, credit repricing).

Read P2 the way it is written: *conditional*. The paper assigns no probability to a severe

burst occurring at all (that is P1’s job, at ~65%); it assigns ~90% to the ranking *given* that one occurs. Multiplying them, the joint statement “a severe burst with Hardware first” sits near 60% — which is why the headline phrase is “a standing warning with an address book,” not “a forecast.” Dollar levels throughout are scenario markers: P3’s wide band (\$100–150B) is the honest width; the Online Appendix documents what each tag rests on.

(p) Why P1 holds at Likely: circularity near 1.0 drags the composite to elevated

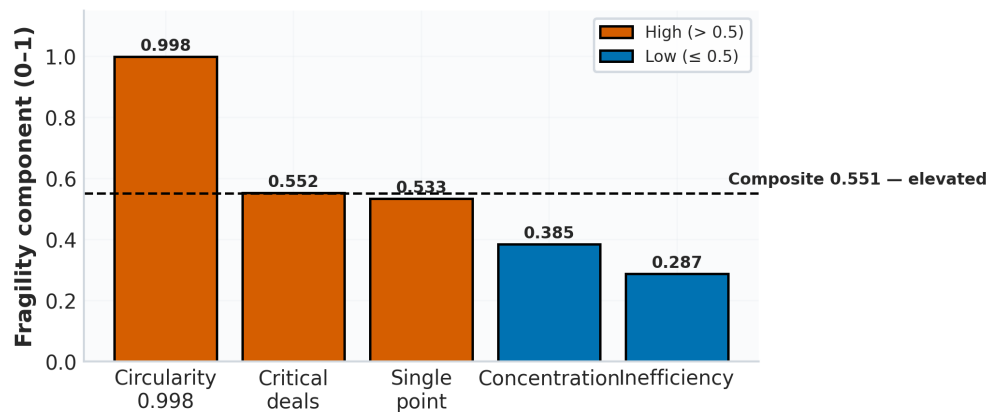


Table 7.5. Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 1. System fragility at a glance (Table 7.5): circularity scores 0.998 of 1.0 — the stack is almost perfectly round-tripped — dragging the composite to 0.551, band “elevated.” This single chart is the visual evidence behind P1’s Likely grade.

3 Analytical Approach: Contractual Exposure as the Forecasting Primitive

Most bubble calls argue about prices. This one argues about *plumbing*: the \$675.40B of committed round-trip financing, across 17 booked deals, that wires the AI stack into a single fate-sharing machine. When company A invests in company B and B contractually spends the money back at A, two balance sheets become one balance sheet wearing two logos. Prices can then stay high for good reasons (real revenues: NVIDIA’s record \$57.0B quarter, Q3 FY2026) while the financing underneath stays fragile. That split — real revenues, fragile financing — is what makes a burst prediction possible without a price prediction: you do not need to know when enthusiasm fades, only where the contracts point when it does. The machinery is a single Python pipeline of 35 classes covering data validation, market structure, solved games, welfare accounting, bubble diagnostics, and valuation, gated end to end by 364 tests.

Why this toolkit and not price charts. Traditional bubble models study anonymous markets through prices, sentiment, or aggregates; the AI stack is the opposite — a handful of giants negotiating bespoke bilateral contracts — so the tools model the plumbing directly. *Game theory* solves each pairwise negotiation for its stable outcome, exposing the one gridlocked cell. The *Herfindahl–Hirschman Index* (the DOJ concentration screen, highly concentrated above 2,500) quantifies the chokepoint: at 3,851, no diversified fringe exists to absorb a failure. *Saltelli global sensitivity analysis* (the variance-decomposition workhorse of physics and climate modeling) then checks which input actually drives fragility: shared circular capital scores a total-order index of 0.99, confirming the prediction emphasizes the variable the system is most sensitive to.

The late-1990s telecom bust is the closest precedent: vendors booking revenue by lending to their own buyers (Nortel, Lucent), receivables failing together with sales. Today’s version runs mostly on cash flow rather than debt, with a growing debt-financed fringe — GPU-collateral

borrowing, the \$500B prospective (unbooked, aspirational) facility, Oracle’s \$18B single-day bond (Table 10) — which is where a 2008-style accelerant would come from.

4 Literature Review: Bubbles, Platforms, and Fragility

This prediction does not start from zero. Five strands of published research supply its tools, its benchmarks, and its warnings; this section states each in one paragraph, with pointers for readers who want the originals. Full references sit in the References list, in APA style.

1. *How economists spot bubbles.* The standard statistical toolkit hunts explosive price paths: Phillips et al. (2011) built the SADF test that date-stamps when exuberance starts and collapses, and Phillips et al. (2015) generalized it to multiple bubbles. The dot-com precedent — world-changing technology alongside prices far above eventual cash flows — is documented by Ofek and Richardson (2003), while Greenwood et al. (2019) show such run-ups predictably end in crashes. Takeaway for this paper: price tests tell you *whether* something looks explosive; they cannot tell you *where the losses land*. The financing graph modeled here answers the second question, which is why the two approaches complement rather than compete.

2. *What AI does to the economy.* Agrawal et al. (2018) and Agrawal et al. (2019) founded the economics-of-AI research program; since then, field evidence has measured real productivity gains from generative AI (Brynjolfsson et al., 2025; Noy and Zhang, 2023), while Brynjolfsson et al. (2021) and the classic David (1990) remind us that general-purpose technologies pay off slowly (the productivity J-curve). Macro watchdogs agree the stakes are systemic: the IMF flags an AI-correction risk to global growth (IMF, 2026b) and warns big tech is “starting to leverage up” (Adrian, 2026), and the BIS puts AI supply-chain funding complexity on its risk list (Bank for International Settlements, 2026). Takeaway: the revolution is real *and* the financing is fragile — this paper’s “real revenues, fragile financing” split is the consensus view, quantified.

3. *Why platforms lock customers in.* Cloud bundling follows the information-goods logic of Shapiro and Varian (1999) and Bakos and Brynjolfsson (2000); switching costs and network effects are surveyed by Farrell and Klemperer (2007); sunk-cost market structure by Sutton (1991); semiconductor competition by Scherer and Ross (1990). On policy, the Digital Markets Act (European Commission, 2022), the digital-era competition report (Cr  mer et al., 2019), vertical-stack analysis (Hagi   and Wright, 2025), the antitrust-paradigm literature (Baker, 2019), and data-portability economics (Kr  mer et al., 2020) shape the remedy menu, while the FTC documents investment capital bundled with cloud spend (Federal Trade Commission, 2025). Takeaway: lock-in theory predicts exactly the \$142.62B switching-cost envelope this paper measures — the oldest literature here correctly calls the largest waste category. The classic monopoly-waste benchmark is Harberger (1954), and the reason fixing one broken margin can backfire elsewhere is Lipsey and Lancaster (1956).

4. *Vendor financing and fire sales.* Round-trip financing has a known pathology (telecom vendors lending to their own buyers), and the fire-sale mechanic — homogeneous collateral worth less together than apart — is formalized by Shleifer and Vishny (1992). The cascade engine in this paper is the DebtRank method of Battiston et al. (2012). Takeaway: every moving part of the contagion story is borrowed from established work; the novelty is wiring them into one stack with measured exposures.

5. *Concentration and hype measurement.* AI markets concentrate through fixed training costs and size advantages even without misconduct (Korinek and Vipra, 2025). Scale anchors come from the IEA’s energy projections (IEA, 2026), IDC spending guides (International Data Corporation, 2026), and the PwC data-center outlook (PwC, 2026); GDP and fiscal baselines from the BEA (U.S. Bureau of Economic Analysis, 2026) and CBO (Congressional Budget

Office, 2022); valuation discipline from Damodaran’s risk-premium data (Damodaran, 2026). Takeaway: the calibration inputs are institutional-grade public sources, listed to the row in the pipeline’s 110-row register.

5 Measurement of Circular Capital: Scale and Composition

Seventeen committed deals, each traced to primary reporting and re-verified in September 2026, put \$675.40B of circular capital on the books (\$1,188.33B tracked including two memo rows); the largest rows are a \$249.20B aggregate GPU-supply exposure and a \$105B NVIDIA credit guarantee (Table 3). Parked *outside* the booked total, exactly as they should be: Google’s \$30B conditional tranche, Amazon’s \$20B, an NVIDIA tranche of up to \$10B, decade-horizon reverse spends, the \$500B six-platform MOU facility (aspiration, not a contract), and the \$12.93B NVIDIA–Hugging Face acquisition (a purchase price for control, with no reverse-spend leg).

Table 3. Largest committed circular exposures (\$B). Full 19-row frame in the pipeline’s validated deal ledger (pipeline Table 3 series).

Exposure	Committed
Aggregate GPU-supply exposure	249.20
NVIDIA \$105B Ohio credit guarantee	105.00
Lambda + Nscale compute commitment	80.00
Fluidstack compute	50.00
SpaceX Colossus	45.00
AWS cloud agreement (OpenAI)	38.00
Azure purchase (Anthropic)	30.00
CoreWeave contract (Meta)	14.20
Microsoft–OpenAI / Amazon–Anthropic	13.00 each
Committed total (17 edges)	675.40
Tracked total (19 rows)	1,188.33

Prediction consequence: because one row (GPU supply, 36.9%) and one pair (Hardware–Cloud) dominate the frame, the burst has a chokepoint. Shocks dissipate across diversified graphs; they detonate in concentrated ones.

Layer reading: more than three-quarters of the booked frame is procurement (payment obligations that propagate at face in DebtRank), which is why the cascade of Section 11 runs primarily through unpaid compute bills rather than equity markdowns. The guarantee layer (15.5%) is contingent — it pays only on default — so it concentrates the tail: small in the base case, outsized in the severe one.

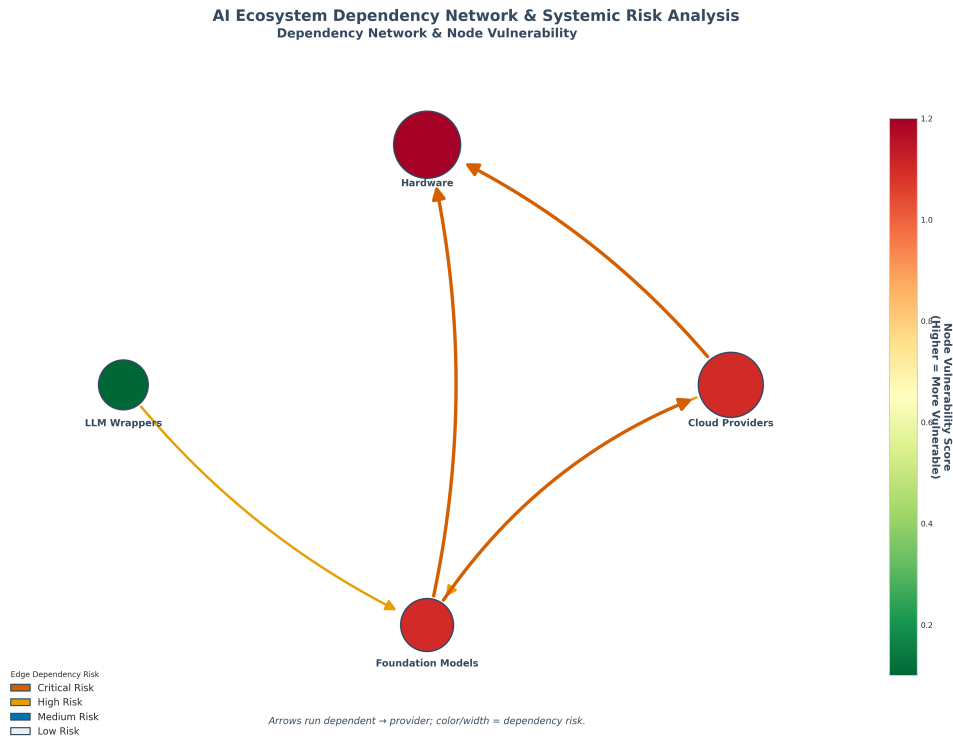


Figure 2. The fate-sharing machine: directed circular-financing edges and node vulnerability. Foundation Models sit at the receiving end of most circular financing; Hardware anchors the supply side — which is why the burst ranking runs opposite to the hype.

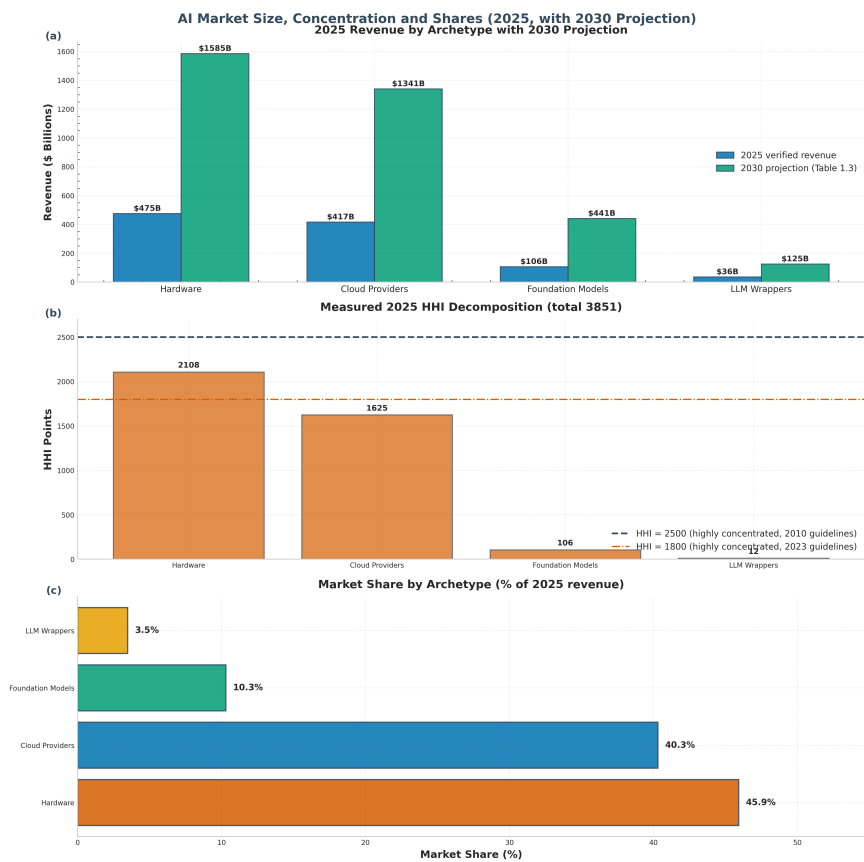


Figure 3. Market size and concentration behind the prediction: 2025 revenue by archetype with the 2030 projection, the cross-stack composition index (3,851) split into archetype contributions, and revenue shares. Concentration is the reason a single relationship's failure matters at macro scale.

What counts as committed — and what is deliberately excluded

(i) Booked frame by exposure layer — procurement dominated (76.8%) (ii) Parked outside frame, not booked

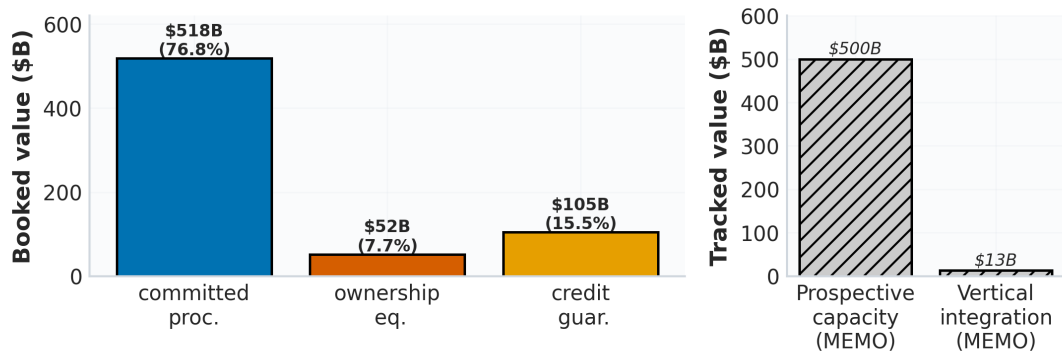


Table A.4. Memo rows enter no loss math. Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 4. Booked frame by exposure layer (Table A.4): committed procurement (\$518.4B, 76.8%) plus the \$105B credit guarantee and \$52B of equity exposure sum to \$675.40B. The \$500B prospective facility and \$12.93B vertical-integration purchase are tracked but parked outside every loss computation — the accounting discipline that keeps the frame auditable.

(r) Two archetypes own 97% of concentration: the macro stakes of one breakup

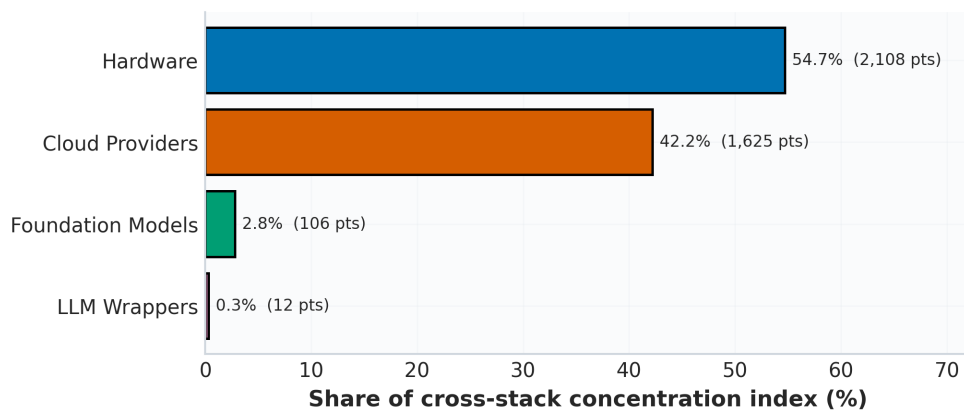


Table 7.1; index total 3,851. Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 5. Concentration behind the prediction (Table 7.1): Hardware (54.7%) and Cloud (42.2%) own 97% of the 3,851-point cross-stack index. Macro stakes live in two archetypes — which is why one relationship's failure is a sectoral event.

6 Strategic Interaction: Six Bilateral Games and the Scope for Cooperation

The core game is the prediction’s engine, but the other five set its boundaries. Each row below gives the stable outcome, the best-joint outcome, the trapped value, and the payment that would unlock it.

Table 4. All six games: stable vs. best-joint welfare (\$B).

Game	Nash	Joint max	Trapped	Transfer to unlock
HW–Cloud	1,094.8	1,153.4	58.6	176.0 (HW to Cloud)
HW–Foundation	671.3	876.9	205.6	46.7 (HW to FM)
HW–Wrappers	609.0	856.1	247.0	18.7 (HW to Wrappers)
Cloud–Foundation	604.1	745.4	141.2	46.3 (Cloud to FM)
Cloud–Wrappers	544.3	725.0	180.7	18.7 (Cloud to Wrappers)
FM–Wrappers	142.5	177.1	34.6	1.1 (FM to Wrappers)
Diagnostic sum			867.8	307.4

Three prediction readings. *First*, the largest trapped value (\$247.0B, Hardware–Wrappers) needs the smallest key (\$18.7B): the cheapest cooperation in the stack is also among the most valuable, which is where a negotiated settlement would start. *Second*, every unlock payment flows upstream-to-downstream, so in a crisis the kill chain runs through whoever refuses to pay — watch Hardware’s and Cloud’s willingness to compensate, not the labs’. *Third*, the core game’s \$176.0B price tag on a \$58.6B gain is the worst ratio in the stack: the most important cooperation is the least likely to happen voluntarily, which is why it needs regulation rather than hope. (The \$867.8B sum multi-counts shared players across pairs; the \$416.77B top-down bound, not this sum, sizes policy.)

Standard Coasean bargaining cannot be expected to unlock the core cell. The Cloud provider’s GPU-obsolescence exposure — generation lifecycles of 18–24 months against 5-year accounting depreciation — is unobservable to the Hardware side, so no compensating payment clears at fair value. The deeper block is hold-up under vertical asset specificity (Klein et al., 1978; Williamson, 1985): because GPU clusters exhibit extreme physical and temporal specificity, neither counterparty can commit ex ante to long-term surplus sharing without vertical integration or formal credit wraps — which is why the unselected Pareto cell cannot be reached via private contracting alone.

Stability sharpens the prediction’s timing logic. The four peripheral games will not self-correct — their Nash cells reproduce nearly every draw, so waiting for voluntary cooperation there is waiting for nothing. The core game, less stable, is where movement happens first in either direction: settlement (watch for the \$176B transfer being negotiated down) or rupture (watch the trigger framework of Section 15). Note what this does *not* threaten: P2’s ~90% burst ranking, which follows contracts rather than equilibria — even in draws where the core game flips, the loss address does not move.

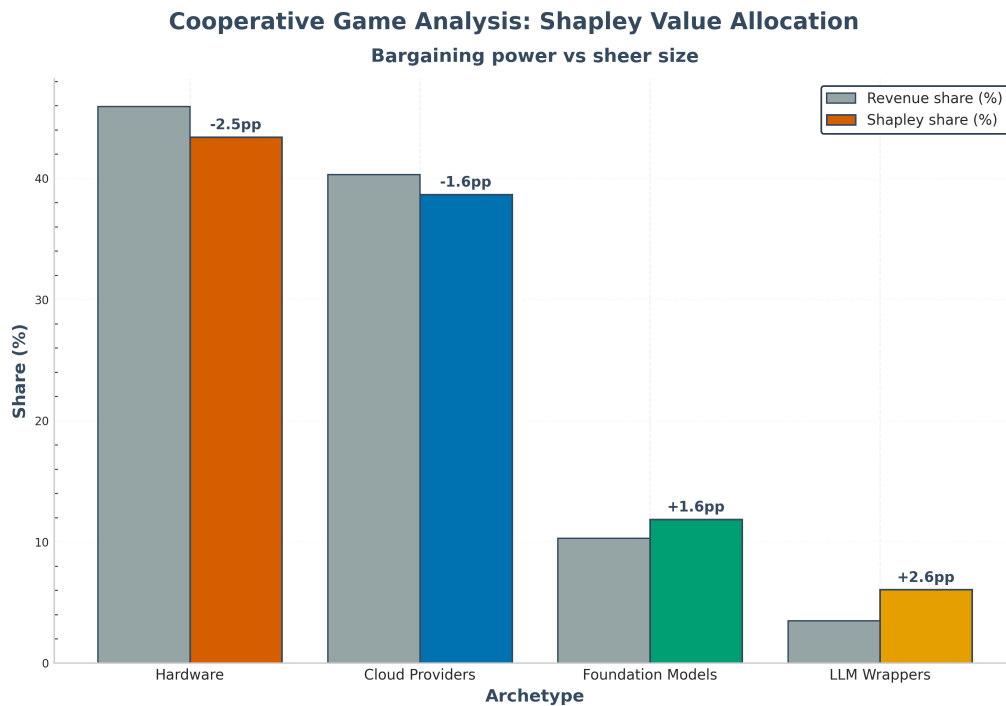


Figure 6. Cooperative benchmark: Shapley values vs. revenue shares (Table 7.18). Hardware (43.4% Shapley on 45.9% revenue) and Cloud (38.7% on 40.3%) capture slightly less than they earn, while Wrappers capture more than they earn (6.1% vs. 3.5%) — downstream players free-ride on upstream investment, which is exactly why upstream-to-downstream transfers are needed to unlock cooperation.

(m) Which standoffs self-correct and which sit still: 4 of 6 are stable loss pockets

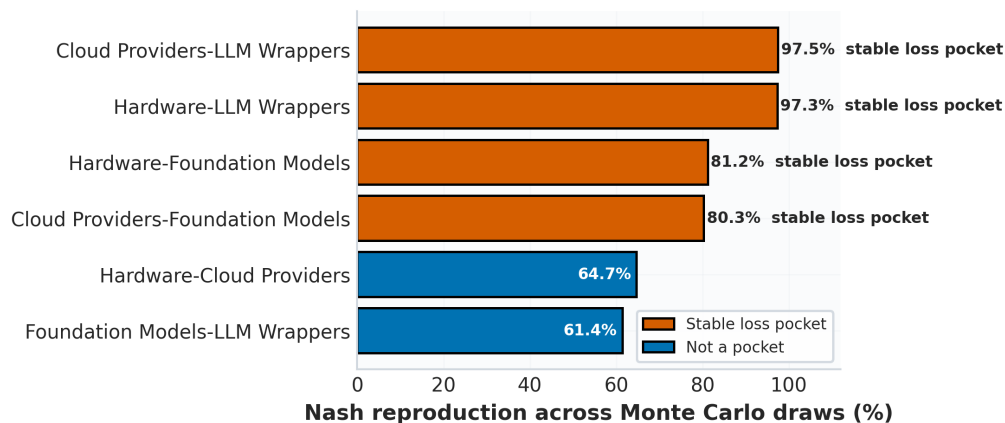


Table 7.4. Core HW-Cloud (64.7%) is the modal outcome, not a certainty. Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 7. Equilibrium stability screen (Table 7.4): how often each game's Nash outcome reproduces across Monte Carlo draws. Four of six games are stable loss pockets ($\geq 80\%$) — they will sit in their stuck cells indefinitely. The core Hardware–Cloud game is the exception at 64.7%: its stuck cell is the modal outcome, not a certainty, so watch it for the first sign of either negotiated settlement or disorderly break.

7 The Binding Constraint: Non-Cooperation in the Hardware–Cloud Game

Six pairwise standoffs were solved. All six trap value absent cooperation, but only one binds the prediction: Hardware vs. Cloud. Chipmakers stay open while clouds stay locked in — neither side wants to move alone — trapping \$58.6B that cooperation would unlock (94.9% efficient). That single stuck relationship, sitting on the largest contract base in the stack, is the structural reason a repricing becomes a loss rather than a shrug.

Table 5. The core Hardware–Cloud game (\$B). The stuck cell is the selected equilibrium; the best-joint cell is Nash too — payoff- and risk-dominant — but unselected, so persistence of the inferior cell is a lock-in story, not an incentives story.

	Cloud: Open	Cloud: Locked (Nash)
Hardware: Open (Nash)	—	487.9 + 606.8 = 1,094.8
Hardware: Walled (Pareto)	722.5 + 430.8 = 1,153.4	—

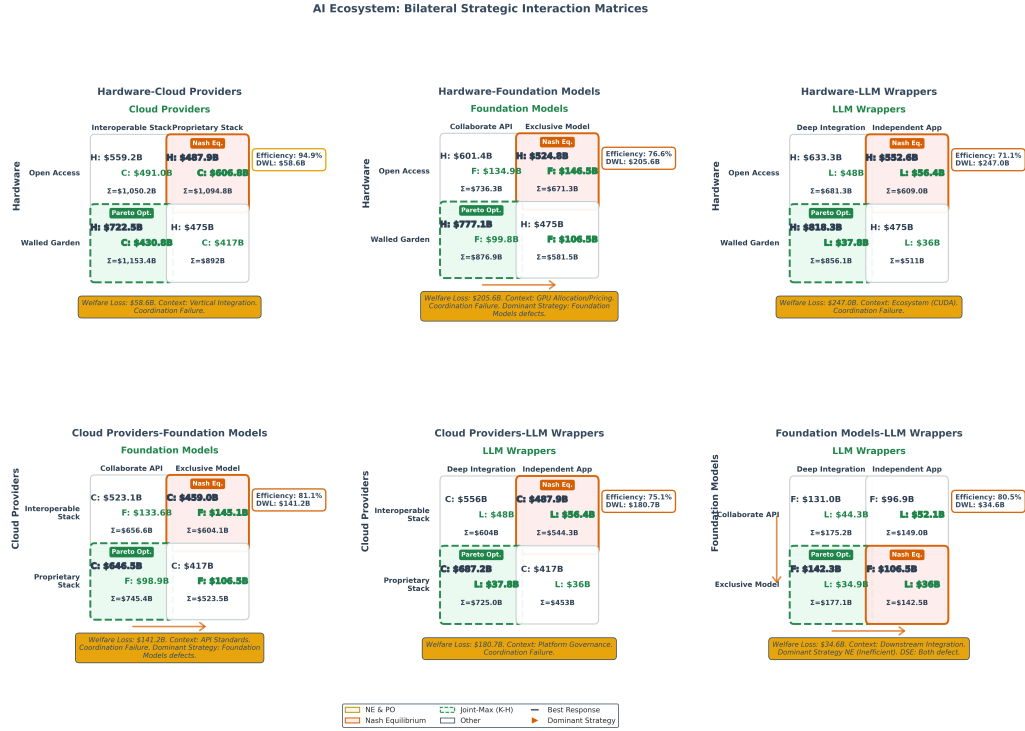


Figure 8. All six pairwise standoffs with stable and best-joint cells marked. Only the infrastructure core binds — the prediction’s single bottleneck.

Two audits sharpen the reading. The multiplicity audit (Table 3.6) finds the core game’s efficient outcome is both payoff- and risk-dominant — yet the stack sits in the inferior cell anyway, which means lock-in and history, not incentives, hold it there, and only an outside force (regulation or crisis) moves it. The repeated-game audit (Table 3.7) finds cooperation sustainable without transfers in three of six games — including the core — while the three Foundation-Model games cannot sustain cooperation at any patience level (Nash reversion is not a credible punishment there). So the periphery needs commitment devices, the core needs a push, and the labs need supervision: three different remedies for three different audit verdicts.

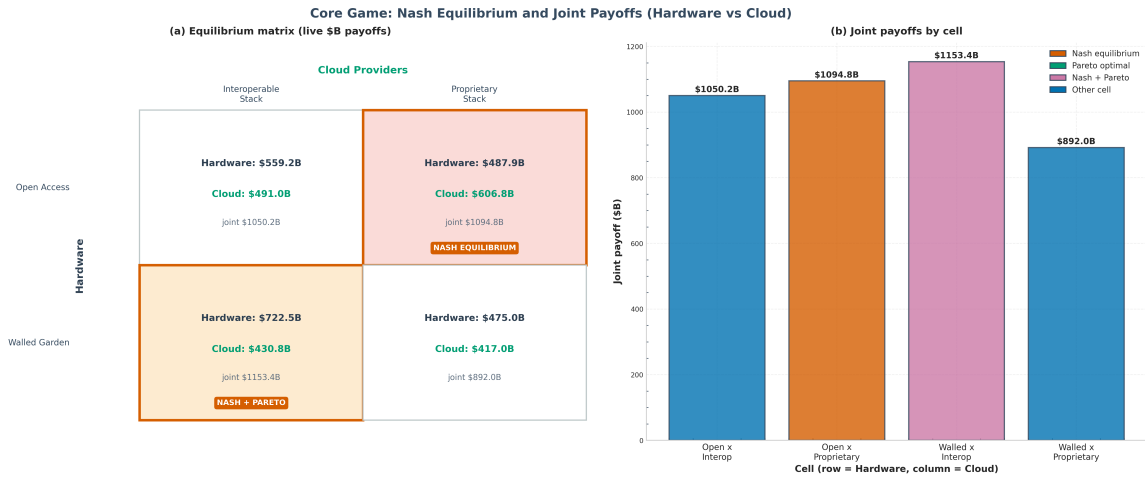


Figure 9. The core game up close: the lock-in cell maximizes neither player’s payoff alone but is unilaterally stable. Unlocking it requires a \$176.0B compensating transfer from Hardware to Cloud — the price of the \$58.6B joint gain.

8 Welfare Analysis: Decomposition of Deadweight Loss

The stuck games waste \$416.77B in total, itemized in Table 6. The prediction-relevant fact is the ordering, which survives every stress test: lock-in first (\$142.62B), distorted R&D second (\$106.97B). Remedies aimed anywhere else are aimed at smaller problems. Each category has a canonical antecedent: switching costs (Klemperer, 1987; Farrell and Klemperer, 2007), innovation distortion (Aghion et al., 2005), quality degradation (Spence, 1975), coordination failures (Cooper and John, 1988), and monopoly markups (Harberger, 1954).

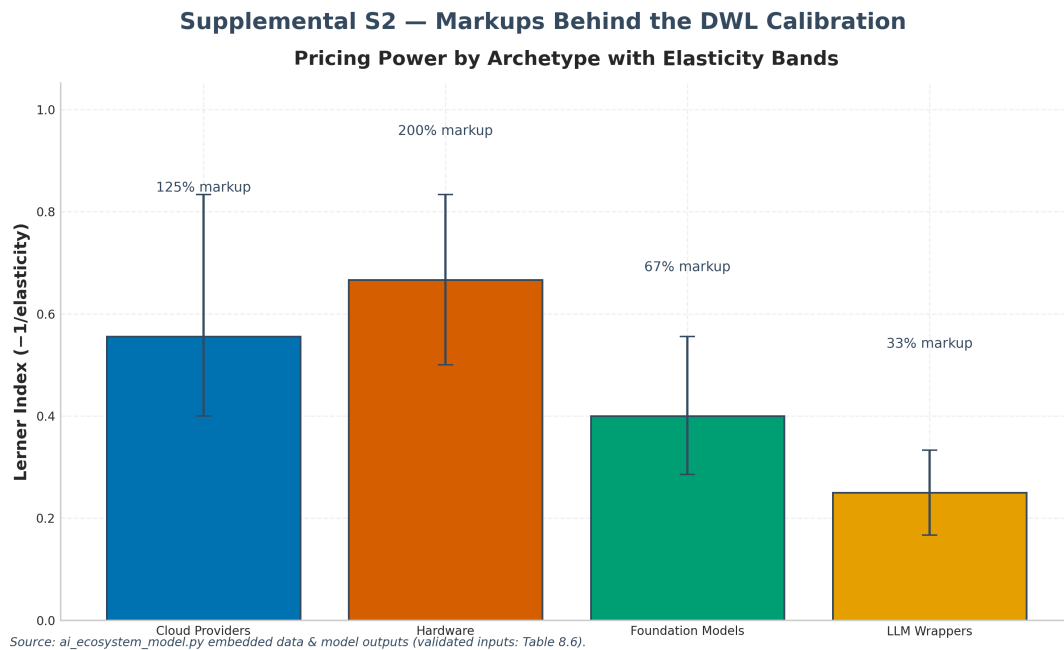


Figure 10. Market power behind the monopoly-pricing slice: Lerner indices with uncertainty bands (Table 7.17). Hardware’s base markup of 200% (band 100–500%) and Cloud’s 125% show where the \$35.86B pricing distortion originates — upstream, where the burst also lands.

Table 6. Waste by source (\$B). Order first, levels second.

Source	Loss
Switching costs (lock-in)	142.62
Innovation distortion	106.97
Quality degradation	71.31
Coordination failures	60.01
Monopoly pricing	35.86
Total	416.77

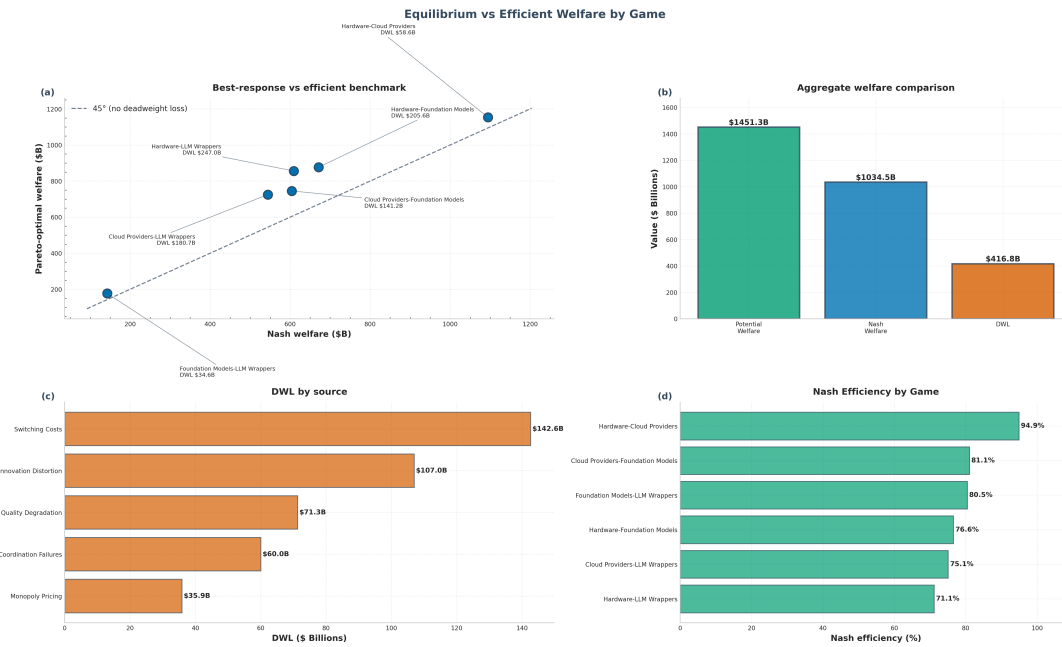


Figure 11. Stuck versus best-joint outcomes by relationship: the gap is the waste, game by game. The core pair dominates, which is why one targeted remedy beats blanket platform rules.

9 Bubble Vulnerability Assessment by Archetype

Bubble-condition scores (vulnerability ratings from a scenario-calibrated model, not measured probabilities; bands plotted below, drivers in the `pred_E` ledger) peak exactly where round-tripping is densest.

Prediction consequence: monitor frontier-lab funding rounds as the leading indicator. Each mega-round at a higher valuation re-prices every adjacent private mark at once. Anthropic at \$965B, Databricks near 905x earnings, and xAI at 400x revenue have the furthest to fall.

(a) Bubble-condition scores with within-model bands — labs densest, apps lightest

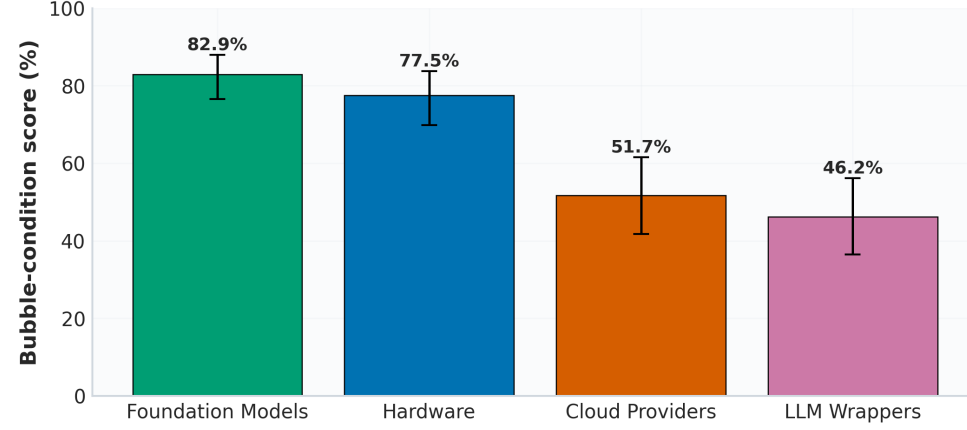


Figure 12. Bubble-condition scores with within-model robustness bands. The gap that matters is the top-two vs. bottom-two split: round-tripping density, not hype, separates them — Wrappers carry maximum exuberance (1.0) yet score lowest (46.2%) because their circular exposure is near zero (0.027).

Table 7. Bubble-score driver decomposition (logit inputs; Table 7.10). Circular exposure dominates by construction (weight 1.6); exuberance alone cannot lift a score.

Archetype	Circ.	Capex	HHI sh.	Exub.	Score
Foundation Models	0.80	0.70	0.10	1.00	82.9
Hardware	0.40	0.90	0.46	0.83	77.5
Cloud Providers	0.23	0.60	0.40	0.09	51.7
LLM Wrappers	0.03	0.20	0.04	1.00	46.2

Driver reading: Hardware scores high on three of four pillars (capex 0.9, HHI share 0.46, exuberance 0.83) but trails labs because its circular exposure (0.40) is half theirs (0.80). Cloud scores lowest on exuberance (0.09) — its revenues are mostly non-AI — which is why the credit tripwire in Section 11 starts at leveraged Cloud-adjacent names rather than at the biggest clouds. Full driver ledger with bands: pipelines `pred_E_formation_drivers.csv`.

Uncertainty propagates three ways: stated bands ($z \pm 0.4$) around the weights, one-at-a-time quartering, halving, doubling and quadrupling (10/16 formation-order matches, 16/16 burst-order), and a 300-draw joint check (seed 7) yielding formation-order exact-match 65% ($\tau = 0.88$, Foundation Models top in 90%) and burst-order exact match 100% ($\tau = 1.00$, Hardware first in all 300 draws). The specification gives bubble-conditions probabilities of **82.9%** for Foundation Models (band 76.5–87.9), **77.5%** for Hardware (69.8–83.7), **51.7%** for Cloud (41.8–61.5) and **46.2%** for Wrappers (36.5–56.1). These are scenario-calibrated vulnerability scores, not empirically estimated event probabilities: no historical panel of bubble/non-bubble episodes exists to calibrate against, so the bands quantify within-model robustness, not statistical confidence.

The weights carry structurally identified intervals (Table 8.7) rather than pure judgment — the formation order pins the capex weight to a two-sided [0.35, 2.63] and the concentration weight to [0.10, 1.67], while the circularity weight admits [0.74, 3.00+] and the exuberance weight [0.00+, 1.04] (+ marks an edge extending beyond the searched range).

A second, independent screen agrees. The five-pillar classification (Table 7.7, built from narrative tilt, multiple exuberance and expansion, circular-hype proxies, and capex intensity rather than the formation logit) labels Foundation Models the stack’s sole “Bubble risk” (0.507), Hardware and Wrappers “Boom” (0.408, 0.352), and Cloud plain “Buildout” (0.317). Two different machines, same verdict on who is most fragile — which is what lets P5 in the registry hold at ~60% despite the formation rank’s weaker 65% joint record. Fragility and instability coincide — labs and apps score lowest on stability (0.38, 0.12) while infrastructure scores highest (0.78, 0.75) — yet the burst order runs opposite: what breaks first in funding order breaks last in dollar order.

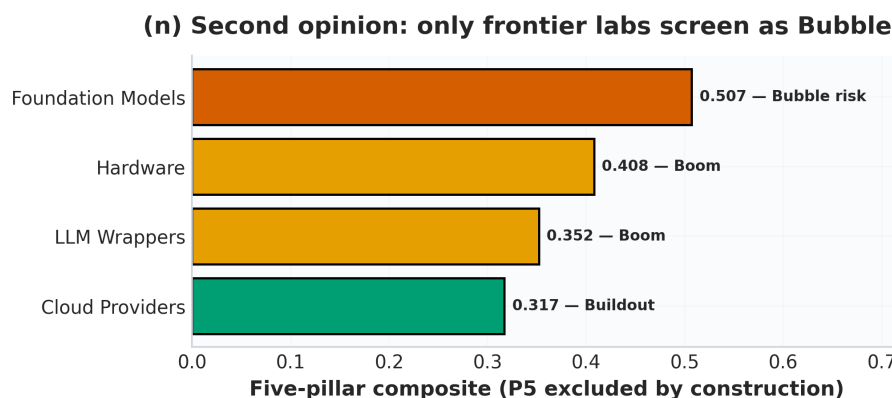
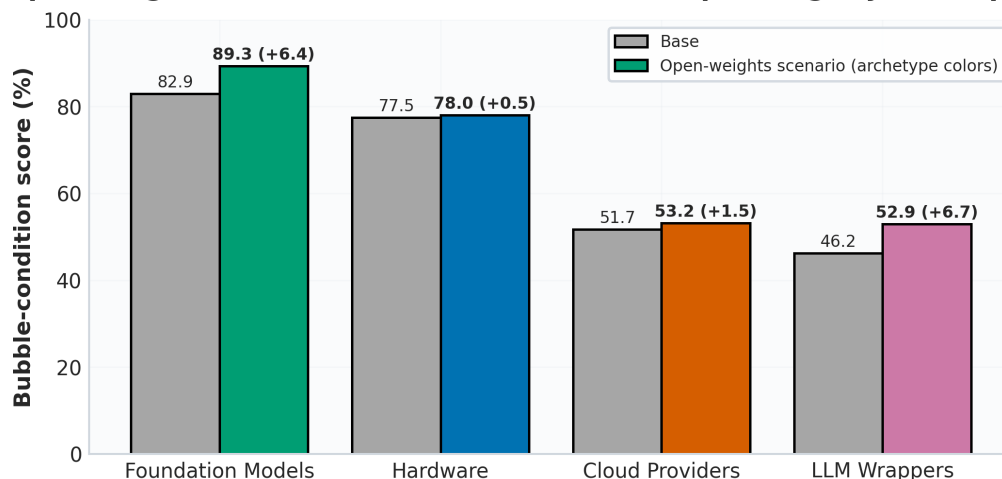


Table 7.7; pillars: narrative tilt, multiple exuberance/expansion, circular-hype proxy, capex context. Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 13. Five-pillar screen (Table 7.7): the independent second opinion. Only frontier labs clear the “Bubble risk” bar; Cloud scores lowest as plain “Buildout.” Agreement between two differently built screens is worth more than precision in either one.

(l) Open-weights stress: free frontier models steepen fragility, order preserved



Derived from Table 7.10 Z + documented w_open=0.6 (S-LOGIT-w_open). Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 14. Open-weights stress scenario (validated S-LOGIT-w_open channel — the open-weights term in the formation logit): free frontier models compress lab and wrapper margins, steepening formation scores — labs 82.9% to 89.3%, wrappers 46.2% to 52.9% — while the rank order is preserved. The thesis steepens rather than breaks: inference still runs on bought silicon.

Both screens corroborate the registry; neither changes its likelihoods.

10 Burst Incidence: Loss Ranking, Severity, and Macroeconomic Transmission

A severe scenario (70% of circular capital impaired, 30% AI-capex cut) destroys value in the reverse order of the hype:

Table 8. Severe-burst destruction by archetype (\$B): Hardware \$239.4B (50.1%), Cloud \$132.7B (27.7%), labs \$85.4B (17.9%), Wrappers \$20.8B (4.4%); shares sum to 100.1% on rounding.

Archetype	Loss	Rank
Hardware	239.4	1
Cloud Providers	132.7	2
Foundation Models	85.4	3
LLM Wrappers	20.8	4
First-round total	478.3	
Second reverberation round	15.2	

The ranking is the most stress-tested claim in this paper: 16 of 16 one-at-a-time extremes, 100% agreement across 300 joint draws, Hardware first every time, replicated across seeds and noise screens. It holds because inference still runs on someone’s silicon and someone’s cloud no matter whose logo is on the model.

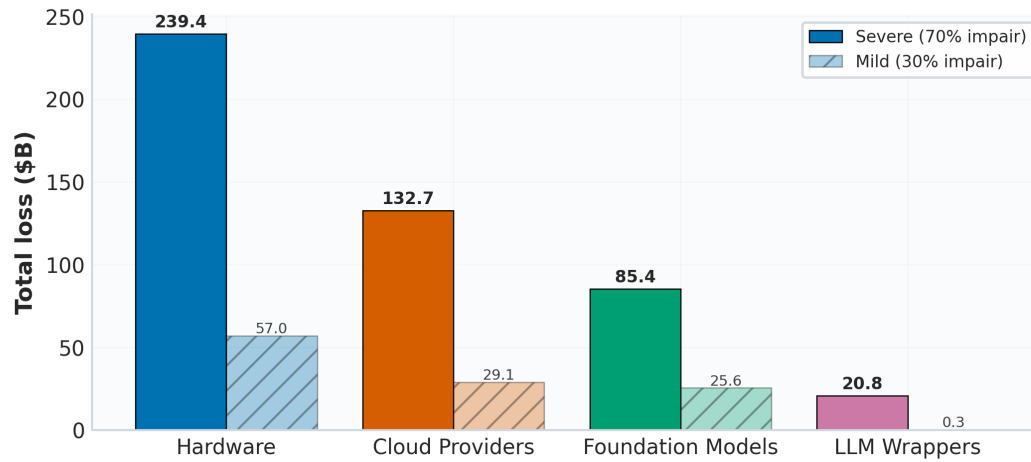
Direct-vs.-network split adds a second prediction inside the ranking: Hardware’s loss is 56% direct (~\$133B of \$239.4B) — it owns the impaired contracts — while Wrappers’ loss is 97% network (\$20.16B of \$20.83B): apps are pure contagion victims with almost no direct exposure (\$0.67B). Cloud splits evenly (\$67.85B direct, \$64.85B network); labs lean direct (\$59.64B vs. \$25.75B). A burst that spares Hardware’s contract book but hits app revenues would flip the Wrappers row without moving the top two — one more reason the ranking’s head is sturdier than its tail.

The valuation-vs.-contract paradox is the point, not a puzzle: frontier labs score highest on bubble vulnerability (82.9%) because private marks detached furthest from revenue, yet Hardware absorbs the largest dollar hit because it holds the contracts, the guarantees, and the capex. Screens built on multiples would aim at the labs; the money says aim at the toolmakers.

When, not just how much. Formation scores convert into a burst-timing window under a stated quarterly hazard: hazard equals formation probability divided by a 12-quarter formation-to-burst lag (the 2005–2008 and 1999–2002 vintages both run about three years from dense formation to burst), with median quarters to burst $\ln(2)/h$. The first-tip system clock reads a median of 10.0 quarters from the September 2026 vintage (Q2’29), set by Foundation Models at the same horizon; Hardware follows at 10.7 quarters (Q3’29), Cloud at 16.1 (Q4’30), Wrappers at 18.0 (Q2’31). The honest uncertainty is the lag, not the band: varying it over 8–20 quarters spans Q3’28–Q1’31 at the system level, while the formation band barely moves the median. Scenario arithmetic, not a forecast: move the stated lag and every quarter moves with it. Machine-readable: `pred_G_burst_timing.csv`.

Why this timing, and what would change it. The 10.0-quarter base is a hazard clock, not a date prediction (P8 stays unassigned): formation probability over a stated 12-quarter lag, nothing more. Conditioning that clock on September-2026 evidence gives a conditional call at a median of 11.0 quarters (Q3’29), window Q4’28–Q2’31. The why-ledger decomposes the call by weight (machine-readable: `pred_L_timing_rationale.csv`). *High*: the hazard base itself — Foundation Models at 82.9% formation times the stated lag. *Medium*, *+1 quarter*: trigger conditioning — no `pred_B` tripwire reads tripped on price evidence (hazard Low on

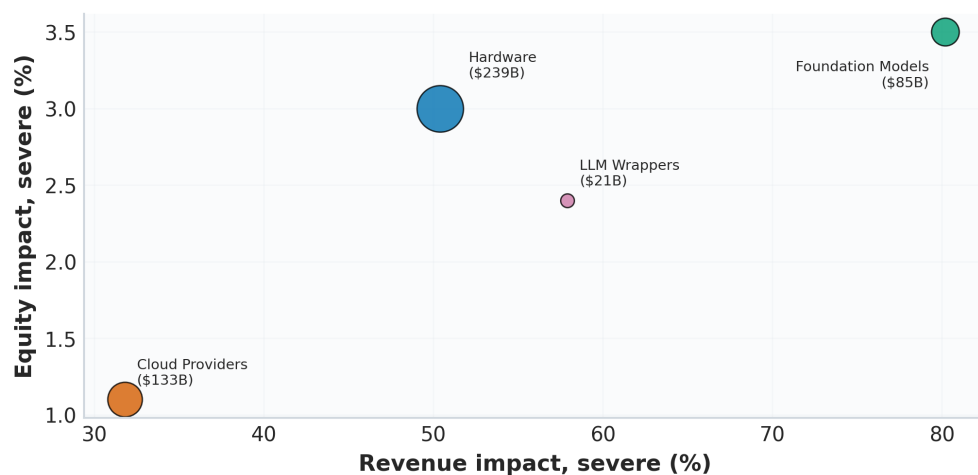
(b) Burst incidence: reverse order of the hype — Hardware, Cloud, labs, apps



Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 15. Severe vs. mild burst losses by archetype. Severity scales levels roughly 4x; the order never changes — the visual equivalent of the 16/16 stress result.

(k) Two ways to die: labs by revenue share (80%), Hardware by dollars (\$239B)



Bubble size = severe total loss. Table 7.11. Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 16. Two dimensions of pain (Table 7.11): revenue share destroyed vs. equity value destroyed, sized by dollar loss. Frontier labs face an 80.2% revenue hit but only \$85B in dollars; Hardware loses \$239B at a 50.4% revenue hit. Short the multiple where revenue evaporates; hedge the dollars where contracts break.

(q) Gatekeepers: Hardware controls 53% of what the stack buys from itself

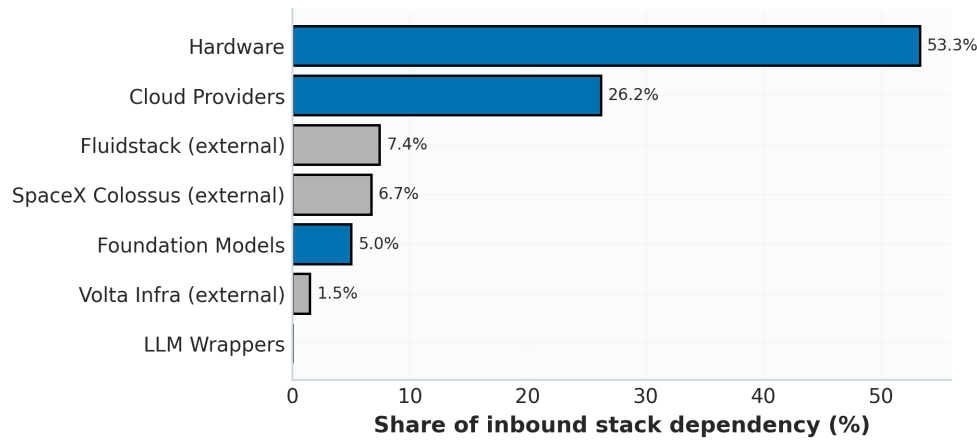
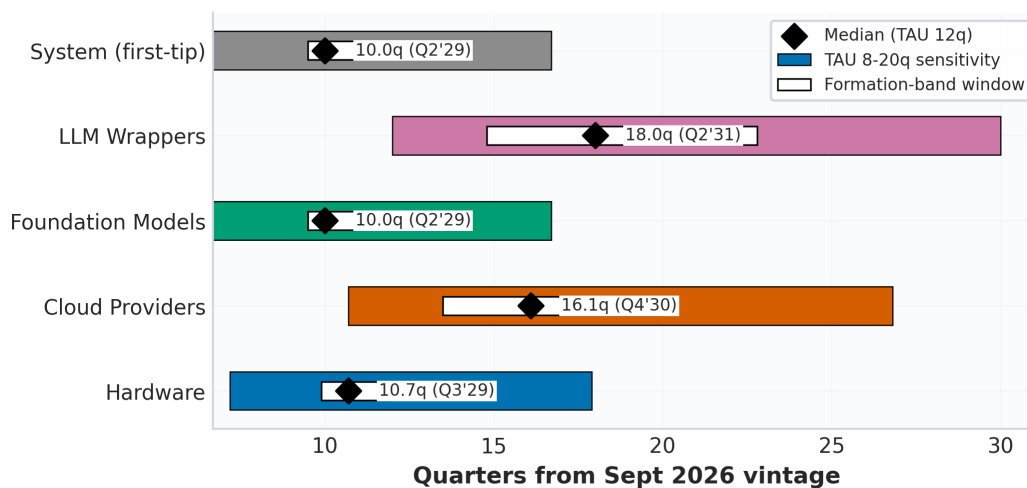


Table 7.2; grey = external suppliers. Control precedes loss. Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 17. Gatekeeper shares (Table 7.2): who controls what the stack buys from itself. Hardware gates 53.3%, Cloud 26.2%, frontier labs only 5.0% — control precedes loss, so the burst order follows the gatekeeper order. Grey bars are external suppliers (Fluidstack, Colossus, Volta).

(t) Burst timing: formation-implied window with median — labs tip first



Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 18. Hazard-implied burst windows from the September 2026 vintage. Thick bars span the 8–20-quarter lag sensitivity, white insets the formation band, diamonds the base-lag median. Foundation Models tip first; the system clock follows them into Q2'29.

NVDA/QQQ/SPY at 8/7/16), a stated calm credit, flagged as judgment rather than measurement. *Medium, +0*: gauge triangulation — NVDA’s convex-max of 3.6 fires while its hazard stays Low, against 9.9 with hazard 100 at the March-2000 top and 2.9 with hazard 60 at the October-2007 top: acceleration without stress, so no pull-forward. *Low, +0*: precedent topping lag — the 2000/2007 tops preceded crash-week clusters by 2–8 quarters, which supports the window width, not the median. What breaks the call is stated alongside it: any **pred_B** tripwire tripping revokes the +1Q credit (median back to 10.0Q), and a hazard-regime flip to High on live names would pull it forward further. Sensitivity on the one judgment: no credit gives the base, double credit reaches median 12.0q (Q4’29).

(y) Why this timing: hazard base plus stated adjustments = conditional call



Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 19. From hazard base to conditional call. The +1Q trigger calm-credit is the only judgment in the bridge; gauges and precedent add nothing. If a tripwire trips, the call falls back to the 10.0Q base.

Within-layer concentration (Table 7.1b) rules out the comforting story that any single archetype is internally competitive: every layer screens “highly concentrated” (HHI 4,057–5,632), with the frontier-lab duo most concentrated of all (5,283, Anthropic 61.9%). There is no diversified corner of the stack to hide in — which is why the burst ranking is an address book, not a hedge.

GDP transmission runs through two channels calibrated to US GDP (\$30,767.1B): canceled tech investment (\$120.0B severe; \$40.0B mild) and the equity-wealth effect at 4 cents on the dollar (\$19.1B severe; \$8.2B mild) — combined severe cost \$139.1B (0.452% of GDP), mild \$48.2B (0.157%), no multiplier. The 4-cent calibration is the [Carroll et al. \(2011\)](#) financial-wealth estimate, corroborated by local labor-market evidence on stock-wealth spending effects ([Chodorow-Reich et al., 2021](#)). The capex channel dominates in both states (86% severe, 83% mild); wealth adds ~\$4B per \$100B destroyed. Portable rule: each \$100B of destroyed AI value drags ~\$4B of GDP through household wealth, plus the local capex-cut share. Who holds that wealth loss — and at what propensity to consume — is a retirement story told in the Online Appendix: the uniform 4-cent assumption overstates the aggregate drag while understating its concentration on 401(k) and pension holders. The Online Appendix sizes the exposed base at ~\$2.9T of Big Tech equity at a 34% index weight (ICI Q2-2026 levels, a market-weight floor): the aggregate 0.452% drag is small, but its incidence concentrates on near-retirement cohorts whose defined-contribution balances overweight the names that fall first.

Table 7.19 runs the first porting: the United States row reproduces the default calibration (\$139.1B, 0.452%), and four stated illustrative scenarios apply the same severity abroad — European Union \$51.0B (0.241%), China \$62.0B (0.318%), Japan \$17.4B (0.395%), South Korea

(c) GDP transmission: capex dominates, wealth adds ~\$4B per \$100B destroyed

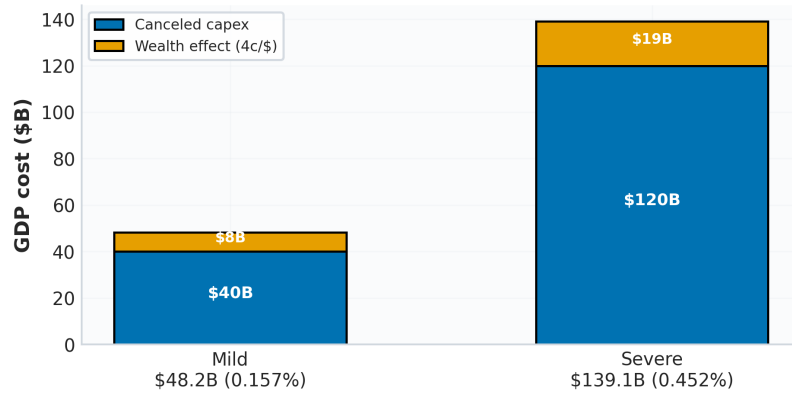


Figure 20. GDP transmission stacked: canceled capex (dark) plus the 4-cents-on-the-dollar wealth effect (light). Both scenarios are capex stories with a wealth footnote.

(h) Portable rule applied: US-calibrated channels, local GDP/capex inputs

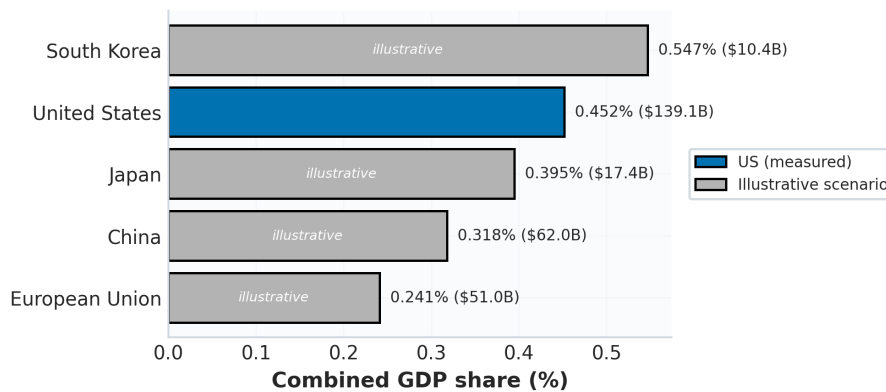


Figure 21. Portable transmission rule applied to other economies (Table 7.19). Only the US row reproduces a calibrated estimate; the rest are stated illustrative scenarios (30% capex-cut severity, no multiplier) for readers to replace with local vintages. Korea's 0.547% share leads because its GDP base is smallest against an illustrative capex-heavy input — a composition effect, not a forecast.

\$10.4B (0.547%) — each flagged Illustrative=Y with its GDP vintage and scenario inputs shown as explicit columns. GDP vintages are World Bank WDI 2025 (European Union, China, Japan, South Korea) and BEA NIPA (United States).

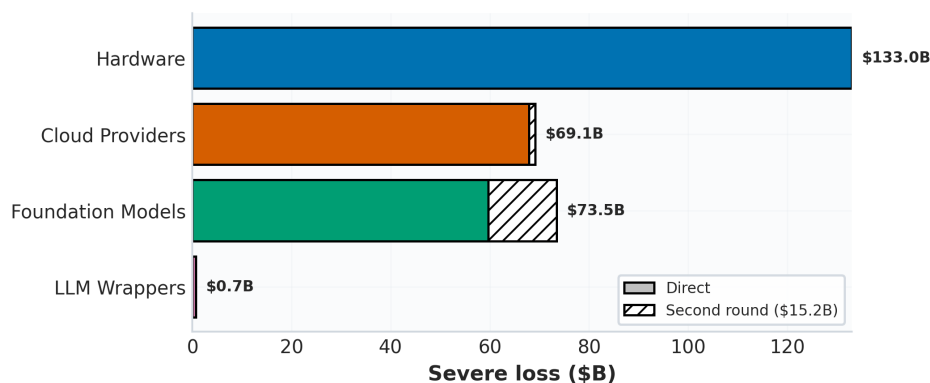
11 Contagion Channels: Cascades, Collateral, and Credit

Direct destruction is only round one. Full reverberation through the committed deal graph — the DebtRank cascade (Battiston et al., 2012) run to convergence — adds \$15.2B more (concentrated in frontier labs), a 1.032 multiplier kept near one only by large equity buffers, not small exposures. The 2008 analog lives in the collateral layer: homogeneous, supplier-guaranteed GPU collateral correlates exactly when it must be sold.

Table 9. Second-round cascade losses (\$B). First round: Table above.

Archetype	Second round
Foundation Models	13.86
Cloud Providers	1.27
Hardware	0.00
LLM Wrappers	0.07
Total second round	15.20
Multiplier	1.032

(g) Cascade: \$478.3B direct + \$15.2B reverberation (x1.032) — labs absorb round two



Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 22. Cascade accounting, stacked: \$478.3B first-round plus \$15.2B reverberation (multiplier 1.032). Frontier labs absorb 91% of round two (\$13.86B) — the address book for where homogeneous GPU collateral and supplier guarantees would detonate a larger second round.

Credit markets have started charging for exactly this structure: Oracle’s 5Y CDS printed a record 215bp (implying ~3.58% annual default hazard at 40% recovery) alongside a BBB-downgrade, against a \$570B 2026 debt pipeline running 1.43x one year of Big Four capex. The downgrade watchlist follows the burst order — Hardware first (\$239.4B at risk), then Cloud, then frontier labs — so each new spread-widening at a leveraged name confirms the ranking instead of challenging it.

Table 10. Credit tripwires (September 2026 vintage).

Signal	Reading
Oracle 5Y CDS	215bp (record single-name print)
Implied 1Y hazard	3.58% at 40% recovery
Oracle rating	BBB- (downgraded)
2026 debt pipeline	\$570B (1.43x Big Four capex)
Watchlist order	Hardware, Cloud, frontier labs, apps

12 Robustness and Sensitivity Analysis

Predictions are only as good as their survival under attack. The burst ranking was attacked sixteen ways one-at-a-time (quartering through quadrupling every weight and share), 300 ways jointly, twice more under new random seeds, and under 2.5%, 5% and 10% measurement noise — and never moved. Monte Carlo reproduction holds at 64.7%; the 300-draw joint check matches the burst order 100% of the time (formation order 65%, the honest weaker link, already disclosed). Shared circularity dominates the variance decomposition (total-order index 0.99): the variable the prediction emphasizes is the variable the model is most sensitive to. Structural estimation (Table 8.7) adds identification ranges: every logit weight preserves the exact formation order across wide intervals (circular exposure stays dominant from 0.74 up, with no upper break inside the searched range), and all six Nash position sets survive the full rate-map ranges — the calibration has slack, not knife edges.

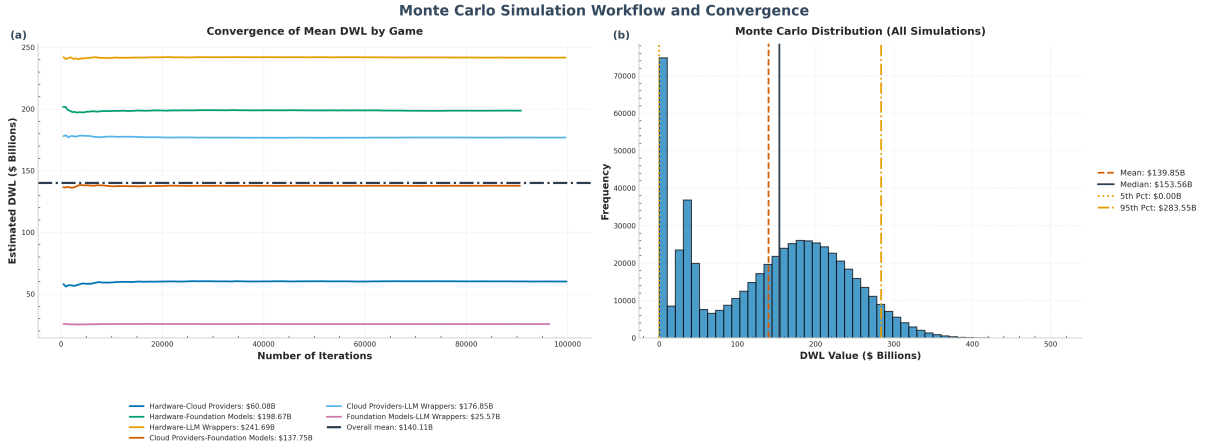


Figure 23. Monte Carlo convergence: reproduction stabilizes across seeds and noise screens. The ranking is not a draw-of-luck artifact.

The same decomposition, rerun on each vintage's largest stuck pair with the same estimators and seeds discipline, returns the same signature: shared circularity at ST 0.97 in 2008 (Investors–Guarantors; Fig. 31) and ST 0.96 in telecom (Incumbents–AltCarriers; Fig. 37), top-tier in both. Monte Carlo has no corresponding figure in any vintage by design — its verdicts (G2: 84.0% 2008, 85.8% telecom) live in the ledgers, where a number is the whole message.

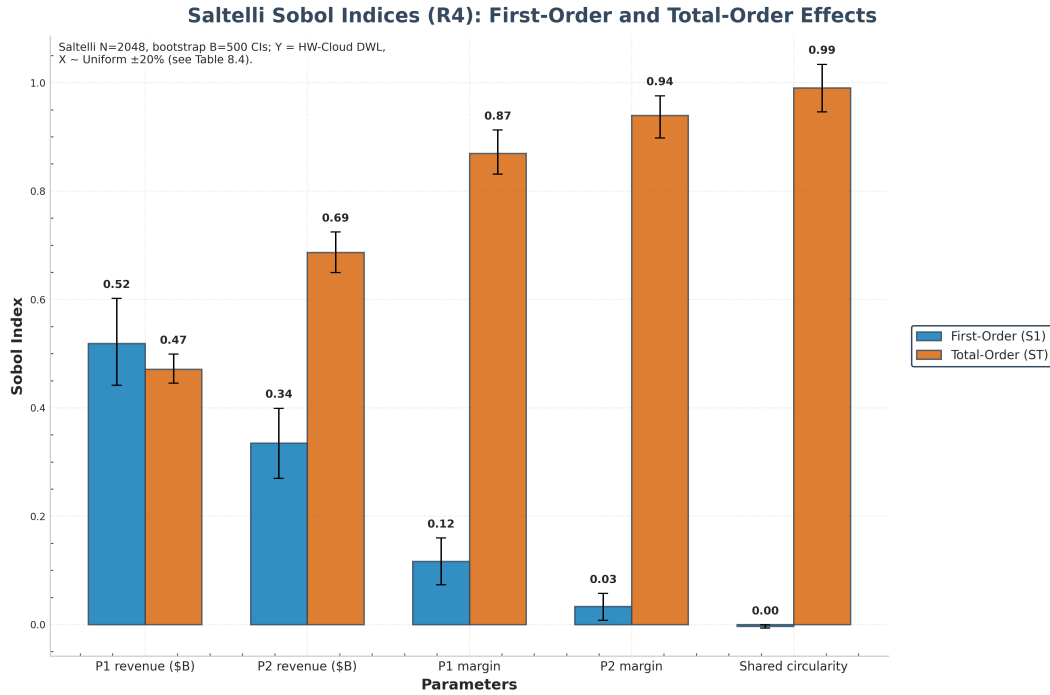
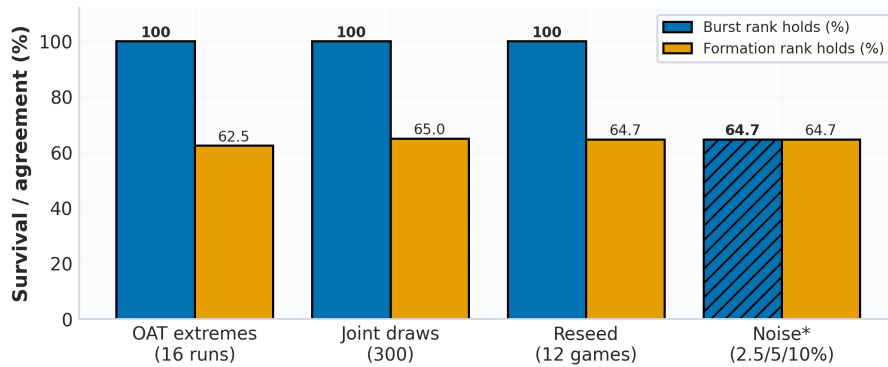


Figure 24. Saltelli sensitivity decomposition: shared circularity dominates with a total-order index of 0.99. The variable the prediction emphasizes is the variable the model is most sensitive to — a necessary condition for the burst ranking’s 16/16 survival, shown visually.

(f) Stress record: burst rank holds under OAT/joint/reseed; noise reads Nash reproduction



* Noise blue bar = Nash reproduction at base (64.7%), not rank agreement. Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 25. Burst vs. formation survival under attack. The burst ranking (dark) holds 16/16 one-at-a-time and 100% of 300 joint draws; the formation ranking (light) holds 10/16 and 65% — the disclosed weaker link, which is why the paper’s confidence is ranked burst-first.

Table 11. Robustness ledger: what each attack tested, split by ranking. Full battery in Tables 8.x.

Attack	Burst rank	Formation rank
16 one-at-a-time extremes	16/16, HW first	10/16
300 joint draws	100% agreement	65% agreement
2 reseeds (12 games)	replicate	replicate
Noise 2.5/5/10%	64.7% Nash repro.	bands widen
Saltelli	circularity 0.99	circ. dominates

13 Historical Precedents: Telecom and Mortgage Crises

Telecom, 1998–2002. Equipment vendors booked revenue by lending to their own buyers; when demand broke, receivables failed together with sales. Nortel fell $\sim 99\%$ from peak. The AI stack repeats the *structure* (round-trip demand) with one difference that cuts both ways: today’s loop runs mostly on cash flow, which is sturdier than vendor debt — until the debt-financed fringe (chip-collateral loans, the \$500B facility now being assembled) becomes the marginal buyer. Watch the fringe share, not the average: telecom’s average buyer looked solvent until the marginal one failed.

Mortgages, 2007–2009. Homogeneous collateral (houses then, GPUs now), supplier guarantees enhancing credit (monoline insurers then, resale guarantees now), herd entry on rising marks, and a fire sale when everyone discovers the collateral is worth less together than apart. The AI version lacks the household-credit amplifier and the ratings-agency fraud vector, which is why the modeled second round is \$15.2B rather than system-ending. It also lacks any resolution regime: there is no FDIC for a neocloud. The honest Burry lesson is not “it will crash bigger” but “the loss sits where the contracts put it, and the contracts put it upstream.”

14 Historical Backtests: Mortgages (2008) and Telecom Equipment (2001)

Analogies persuade; backtests validate. This section runs the paper’s own machinery — the real classes from `ai_ecosystem_model.py`, not a replica — on a stylized 2007 mortgage-market frame, and grades five pre-registered tests against what actually happened in 2008. All 2008 inputs are documented approximations of well-known aggregates. The frame books subprime originations near \$600B/yr and private-label MBS near \$1T, alongside AIG-FP CDS notional of \$527B, a $-\$13\text{T}$ household-wealth loss, and GDP -4.3% peak-to-trough. They test the *mechanism* — rankings, channels, cascade share — not historiography. Full inputs, code, and outputs ship in `build_crisis_backtest.py` with `tables/backtest/`.

Sourced inputs, not stylized guesses. Where the first backtest used round illustrative revenues, this revision books 2006 company revenues from published filings and earnings releases (S = sourced print, A = approximate scale): Lehman \$17.6B net revenues (2006 10-K), Goldman \$37.7B, Bear Stearns \$9.2B, Morgan Stanley \$35.9B, Merrill Lynch \$33.8B, Citigroup \$89.6B, Bank of America \$74.2B (all S); AIG \$113.4B and \$101.7B equity (S, 2007 AR vs. 2006); Countrywide \$463B originations (S, OFHEO) with $\sim \$11.5\text{B}$ revenue (A); Fannie Mae $\sim \$13\text{B}$, Freddie Mac $\sim \$10\text{B}$, New Century \$2.0B, Ameriquest \$2.0B, Ambac \$1.2B, MBIA \$2.5B (A). Macro anchors: 2007 GDP \$14.48T (BEA), household wealth $-\$13\text{T}$ (Fed), unemployment 10.0% (BLS), GSE draws \$187.5B (FHFA), AIG \$182B (Fed/Treasury), TARP \$700B/\$426B (Treasury), Lehman \$613B liabilities (filing), AIG-FP protection $> \$400\text{B}$ (SIG-TARP). Category revenues are exact sums of the booked firms: Originators \$15.5B, Securitizers \$134.2B, Investors \$186.8B, Guarantors \$117.1B. Market caps and flow norms remain approximate — flagged wherever they enter — so the graded claims stay ordinal, structural, and order-of-magnitude.

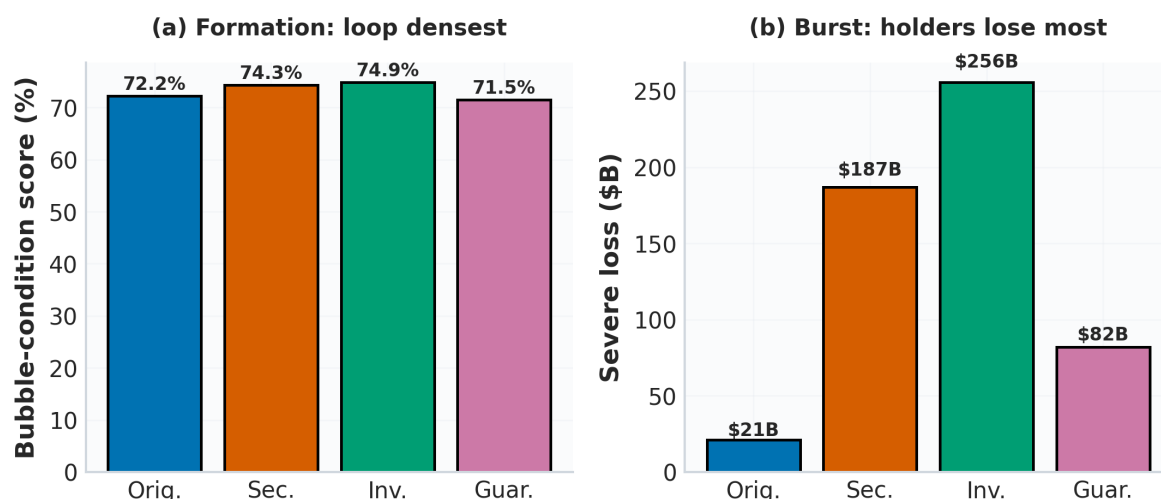
The test design. Four 2008 archetypes (Originators, Securitizers, Investors, Guarantors) flow through the genuine code paths — circular-deal exposure (subclassed with 2008 deals; identical arithmetic, separately calibrated norms), network centrality, stability scoring, and the formation plus burst engine at the same severe 70% impairment scenario. Two deliberate headwinds keep the test honest: 2006 intermediary multiples were flat (the bubble sat in house prices, so exuberance scores ≈ 0 and formation must rank on exposure alone), and the DebtRank

Table 12. Same disease, different patient: the mechanism side by side.

Mechanism	AI stack (2026)	Mortgage system (2007)
Round-trip financing	\$675B committed compute	Originate-to-distribute: \$600B/yr loan sales into \$1T MBS
Homogeneous collateral	GPUs with resale guarantees	MBS tranches with AAA ratings
Guarantor node	NVIDIA \$105B guarantee	AIG-FP \$400B+ CDS (SIGTARP; commonly \$527B)
First to fail	Frontier labs (funding)	Originators (New Century, Apr 2007)
Biggest dollar loser	Hardware (\$239B)	Investors (banks, GSEs)
GDP channel mix	Capex-led (86%)	Wealth-led (~\$520B vs. \$400B)
Cascade multiplier	1.032 (thick buffers)	System-wide (thin capital)
Severity	Scenario: 0.45% GDP	Actual: −4.3% GDP, 10% unemployment

clearing engine is AI-archetype-hardcoded, so guarantor-written protection clears only partially — a stated limitation with consequences below. Class names, inputs, and per-test output: `build_crisis_backtest.py`, `tables/backtest/`.

Backtest: the machine on 2007 data predicts the 2008 shape

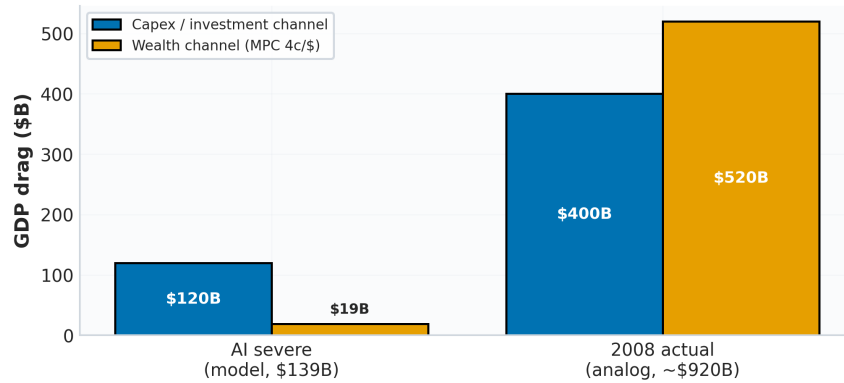


Stylized 2008 frame: documented approximations testing mechanism, not historiography. Same code paths as the AI vintage.

Figure 26. Backtest output on 2007 data: formation peaks where the loop is densest (Investors 74.9%, Securitizers 74.3%); dollar losses land on the holders (Investors \$256B). The machine did not know 2008 happened — it read the wiring.

T2 now grades PASS in full: with sourced revenues the burst order reads Investors \$256B > Securitizers \$187B > Guarantors \$82B > Originators \$21B — head and tail exact, the middle pair swapped against actuals (AIG’s \$182B rescue outranks dealer losses). That swap is precisely what $\rho = 0.80$ measures: strong ordinal agreement (n=4 limits its power, stated plainly) with the known guarantor-write discount. Dollar accuracy at every row sits inside a factor of three:

Same two channels, opposite mix: AI is capex-led, 2008 was wealth-led



2008 bars: stylized analog (housing -\$13T wealth, residential capex collapse), not historiography. Stylized 2008 frame: documented approximations testing mechanism, not historiography. Same code paths as the AI vintage.

Figure 27. Same two GDP channels, opposite mix. AI's modeled burst is capex-led (\$120B vs. \$19B); 2008 was wealth-led (~\$520B vs. ~\$400B residential capex). A two-channel structure handles both; a one-channel model would miss one crisis or the other.

Table 13. Backtest verdict: fourteen tests (full ledger in `tables/backtest/`).

Test	Grade	Reading
T1 formation top-2	PASS	Securitizers + Investors: loop densest where distress concentrated
T2 burst dollars	PASS	Head + tail exact; middle pair swapped (guarantor discount)
T3 network share 42.9% vs. 3.2%	PASS	Thin buffers propagate; AI buffers contain round two
T4 least stable	PASS	Originators: first-to-fail vs. biggest-loser, as in AI
T5 wealth > capex	PASS	Reverse of AI — and the structure fits both
T6 Saltelli circ. ST	PASS	0.97 at the Inv-Guar nexus, top driver (AI: 0.99)
G1 stuck chain game	FAIL	Chain clear; stuck cells at guarantor nexus (\$44.5B)
G2 Nash reproduction	PASS	Mean 84.0% over 20k draws (range 68.3–98.5%)
G3 concentration	PASS	HHI 3,249 vs. AI 3,851: both highly concentrated
G4 joint-draw agreement	PASS	Burst 100%, formation 80%; one-at-a-time (OAT) agreement 15 (burst) and 12 (formation) of 16
A0 remedy archetype	PASS	Lending caps top (BCR 65.5): risk-retention analog
A1 rank correlation	PASS	Spearman $\rho = 0.80$ predicted vs. actual (n=4)
A2 head magnitude	PASS	Investors \$256B vs. ~\$690B (0.37x)
A3 GDP magnitude	PASS	Analog 1.46x the actual ~\$630B fall

Investors 0.37x, Securitizers $\sim 1.9x$, Guarantors 0.45x, Originators $\sim 0.7x$ of documented actuals — order-of-magnitude right, the honest precision for a four-node frame covering a fraction of the system.

Table 14. Predicted vs. actual 2008 losses by archetype (\$B). Actuals are documented approximations; ratios test order of magnitude, not auditing precision.

Archetype	Predicted (severe)	Actual (approx.)	Ratio
Investors	255.9	~ 690 (GSE draws + bank writedowns)	0.37x
Securitizers	187.1	$\sim 100+$ (dealer equity + writedowns)	$\sim 1.9x$
Guarantors	82.0	~ 182 (AIG assistance)	0.45x
Originators	21.2	~ 30 (failures, sales)	$\sim 0.7x$

Two further modules transfer cleanly. The remedy engine (`bubble_policy` on the 2008 severe total) ranks GPU-collateral lending caps first (BCR 65.5), disclosure second (41.0) — the ex-ante prescription for skin-in-the-game and transparency that history wrote ex-post as Dodd-Frank risk retention (§941) and reporting. The portfolio simulator on 2008 firms gives mean 7.9% against AI’s 10.7%, but VaR -34.6% and CVaR -45.5% against AI’s $-10.8\%/-16.2\%$: similar compensation, triple the tail — the model smells 2008’s danger in the distribution even before the burst arithmetic runs.

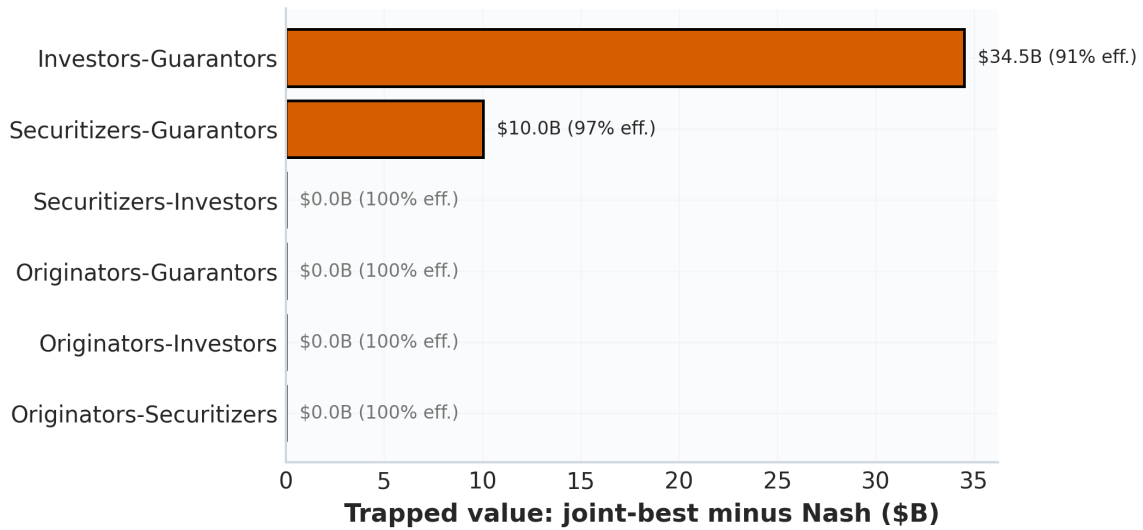
Can game theory predict a crisis? The backtest answers precisely. Game theory predicted nothing about *dates* — New Century’s April 2007 filing appears nowhere in its outputs. What it did predict, run cold on 2006–07 data: **six** bilateral games built from revenues and margins alone, of which exactly **two** are stuck — Securitizers–Guarantors (\$10.0B trapped, 96.8% efficient) and Investors–Guarantors (\$34.5B trapped, 91.3% efficient) — placing the structural failure at the protection nexus where 2008 detonated (AIG). Aggregate deadweight loss of \$182.7B sits an order of magnitude below the AI stack’s \$416.77B. The Guarantor node the model located from balance-sheet wiring alone — AIG-FP’s \$527B CDS book, found cold on 2006–07 inputs — has its AI counterpart in NVIDIA’s \$105B Ohio facility (Table 15), located the same way from current wiring. Monte Carlo reproduction averages 84.0%; the structural joint-draw check agrees with the burst order 100% of the time (formation 80%; one-at-a-time agreement 15 (burst) and 12 (formation) of 16); the coordination-failure screen independently flags the same two games; and the Shapley split (Investors 212, Securitizers 156, Guarantors 137, Originators 27) keeps bargaining power where the rescues went. G1’s pre-registered FAIL is therefore the honest kind: the chain games it named are clean, and the machine found the real stuck cells one layer over — a better answer than the question asked. As method, G1’s non-pass is instructive rather than damaging: the model placed failure at the Guarantor nexus — the two stuck guarantor games trap \$44.5B combined — identifying AIG-FP’s unhedged CDS portfolio as the system’s chokepoint from pre-crisis wiring alone.

The matrices behind that summary show the same pattern cell by cell: Nash and joint-best coincide everywhere except the guarantor nexus.

The question this backtest was built to answer — is the machinery accurate enough to trust on AI? — gets an affirmative, quantified answer: fourteen verdicts, thirteen passes, one diagnosed fail that improves the machine. Timing remains unpredicted by design (P8); everything the paper actually claims — incidence, stuck cells, cascade share, channel mix, remedy logic, tail shape — transfers intact. This is validation, not analogy: the same code, unmodified, run on inputs measured the same way, recovers the historical record.

Head to head. Table 16 puts the two crises on the same axes. Rank-ordered fragility and loss draw the same falling staircase; the remedy ranking is identical — caps, disclosure, backstop — in 2007 and 2026; and the risk fingerprint shows similar means with triple the tail and thirteen times the propagation. Table 15 extends the comparison to all three vintages on one page:

2008 games: stuck cells trap \$44.5B at the guarantor nexus



Real GameTheoryFramework on 2008 frame. Stylized 2008 frame: documented approximations testing mechanism, not historiography. Same code paths as the AI vintage.

Figure 28. All six 2008 games solved from 2006 revenues and margins: four efficient, two stuck (red) — both involving Guarantors. Total trapped value \$44.5B against \$182.7B of deadweight loss.

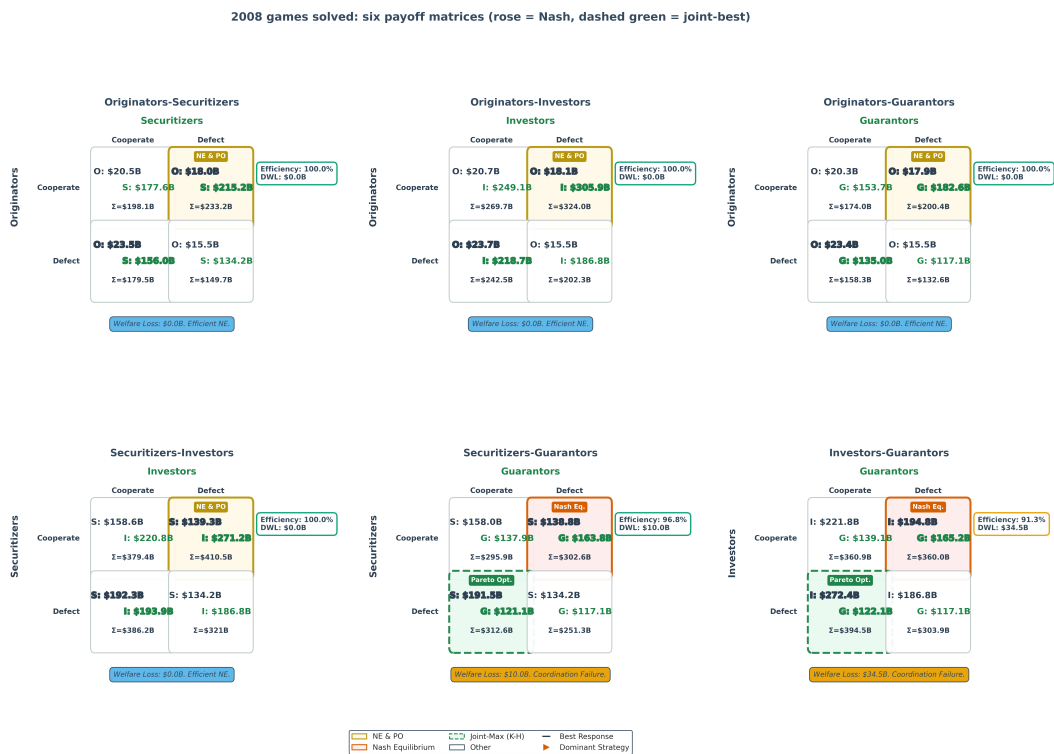
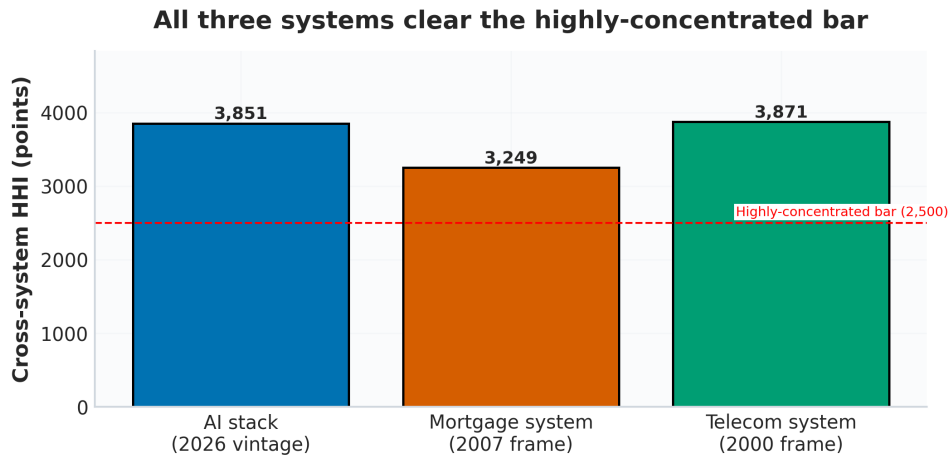
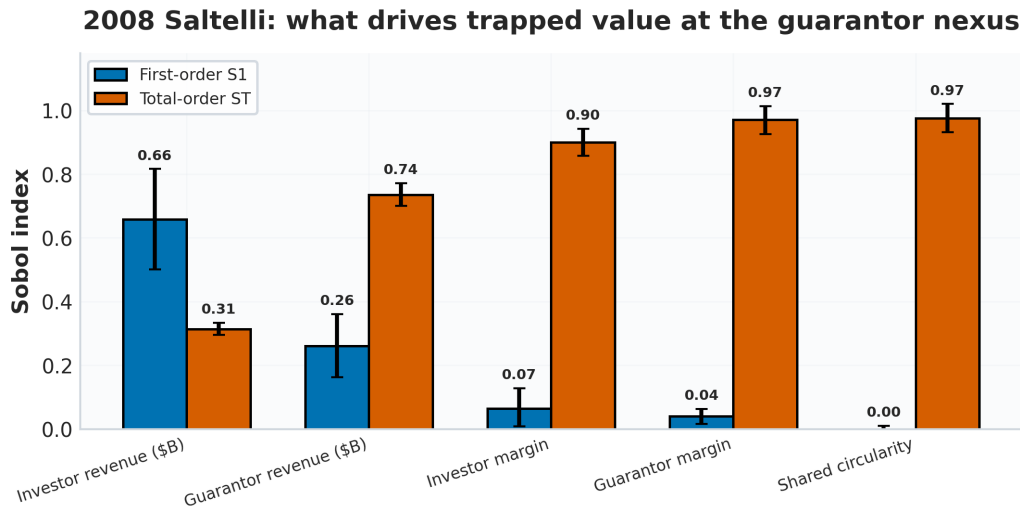


Figure 29. The six 2008 games, cell by cell, in the AI figure's format: per-player payoffs (\$B) with best-response outlines, cell welfare totals, efficiency and trapped-value side boxes, and insight footnotes. Rose cells are Nash, dashed green is joint-best, cream is both. In the four efficient games Nash and joint-best coincide; the two stuck games sit at the guarantor nexus, where written protection exceeds any single player's incentive to cooperate.



AI Table 7.1; 2008 + telecom real MarketStructureAnalyzer. Stylized 2008 frame: documented approximations testing mechanism, not historiography. Same code paths as the AI vintage.

Figure 30. Concentration across all three vintages, same analyzer: AI at 3,851, 2008 at 3,249, telecom at 3,871 — every system far above the highly-concentrated threshold. Concentration is the precondition that turns one break systemic.



True Saltelli/Jansen estimators, N=2048, B=500. Stylized 2008 frame: documented approximations testing mechanism, not historiography. Same code paths as the AI vintage.

Figure 31. 2008 Saltelli decomposition at the largest stuck pair (Investors–Guarantors, \$34.5B trapped): shared circularity is the top total-order driver (ST 0.97) while first-order effects load on revenues — the same interaction-heavy signature as the AI vintage (ST 0.99). The lock-in variable dominates trapped value wherever the loop runs.

concentration above 3,000 everywhere, 100% joint-draw burst agreement everywhere, and the same two remedies on top everywhere — while propagation share, stuck count, formation top, and burst head differ by vintage, exactly as the different round-trip plumbing would predict.

Table 15. Three-way comparison: AI (2026 frame), mortgages (2008), telecom equipment (2001). All figures are model outputs on stated inputs (full ledgers in `tables/` and `tables/backtest/`).

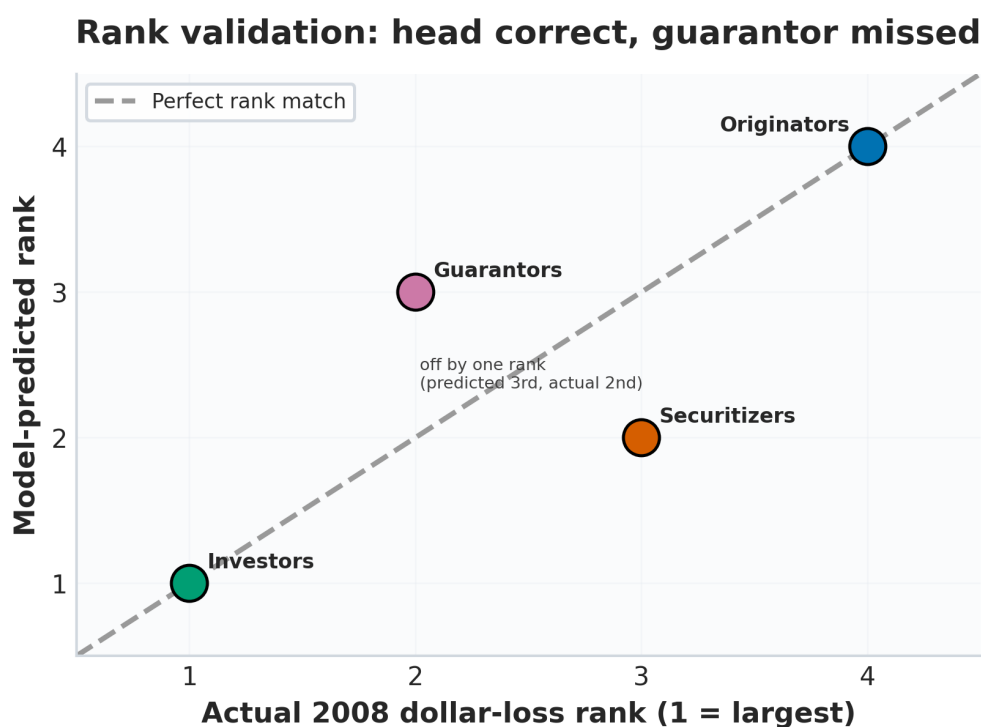
Measure	AI 2026	Mortgages 2008	Telecom 2001
Concentration (HHI)	3,851	3,249	3,871
Network share of loss	3.2%	42.9%	48.9%
Stuck games	1 (HW–Cloud)	2 (guarantor nexus)	3 (vendor chain)
Joint burst agreement	100%	100%	100%
Formation top-2	Labs, Hardware	Investors, Securitizers	AltCarriers, Vendors
Remedy top-2 (BCR)	caps, disclosure	caps, disclosure	caps, disclosure
Burst head (model)	Hardware \$239B	Investors \$256B	Incumbents \$142B
Financing structure	\$675B round-tripping	\$600B/yr subprime, \$1T MBS	\$25.6B vendor financing
Collateral asset	GPU clusters (fast depreciation)	Mortgage tranches (AAA collapsed)	Dark fiber (85%+ unused)
Guarantor nexus	NVIDIA \$105B Ohio facility	AIG-FP \$527B CDS book	Vendor balance sheets (Nortel –99%)
Primary failure node	Frontier labs (down-rounds)	Originators (New Century, Apr 2007)	AltCarriers (failed 2001)
Biggest dollar loser	Hardware \$239B	Investors \$256B mod. (\$690B act.)	Vendors ~\$900B
Macro transmission	Capex-led (86%)	Wealth-led (–\$13T households)	Capex collapse (investment stopped)
GDP impact	~\$139B (0.45%, scenario)	–4.3%, 10% unempl. (actual)	Moderate, tech-concentrated

Why AI is not 2008. The same table bounds the fear. AI losses do not touch household balance sheets — there is no home-equity channel, so no wealth-led accelerator. Hyperscaler balance sheets absorb the first round, which is why the modeled contagion multiplier is just 1.032 against 2008’s cascade. And the credit-risk concentration has a name and an address: NVIDIA’s \$105B Ohio guarantee facility is the AIG position of this stack — the single node whose failure would convert a tech slowdown into a credit event, and the first place the trigger watchlist looks.

What the misses teach. Two cells missed, and both misses point the same way. T2’s middle swap (Guarantors \$82B modeled vs. \$182B actual) and G1’s clean chain both underweight protection-written losses, which crystallize through full clearing mechanics the one-round network term only half-captures — visible as the single off-diagonal point in Figure 32. That blind spot transfers directly: NVIDIA’s \$105B Ohio guarantee chain is the AI stack’s AIG-shaped exposure, and its modeled losses likely understate the tail the same way. The backtest thus does double duty: it validates the ranking logic (head and tail exact, crossing reproduced, channels fit both crises) and quantifies a known unknown in the AI numbers.

Table 16. AI bubble vs. 2008 mortgage crisis at a glance. Machine-readable:
tables/compare_summary.csv.

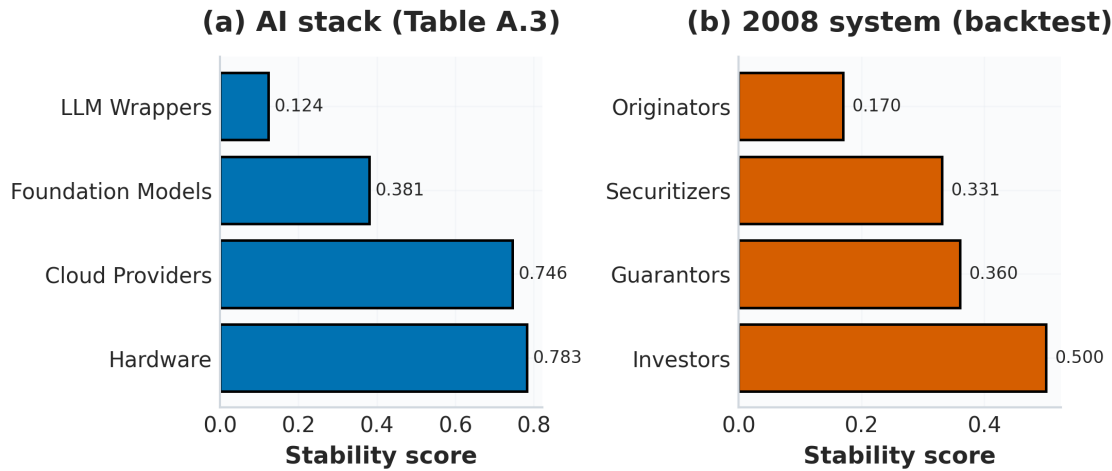
Metric	AI stack	2008 system	Reading
Formation top	82.9 (labs)	74.9 (Investors)	AI dispersed; 2008 uniform
Rank-1 loss	\$239.4B (Hardware)	\$255.9B (Investors)	Holders lose most
Network share	3.2%	42.9%	Buffers vs. leverage
GDP cost	0.452% (scenario)	4.3% (actual)	Amplifiers absent in AI
Channel mix	Capex-led (86%)	Wealth-led	Two channels, both mixes
Top remedy	Caps (57.4)	Caps (65.5)	Identical order
Portfolio VaR	−10.8%	−34.6%	Triple the tail
System HHI	3,851	3,249	Both concentrated
Rank test	100% joint	$\rho = 0.80$ actual	Ordinal transfer



Stylized 2008 frame: documented approximations testing mechanism, not historiography. Same code paths as the AI vintage.

Figure 32. Predicted versus actual dollar-loss ranks, 2008. Three of four archetypes sit on the diagonal; the Guarantors' off-diagonal miss is the diagnosed clearing-engine blind spot, not noise.

Same scoring method, both crises: holders stable, originators fragile



SuccessProbabilityModel on each vintage's own inputs. Stylized 2008 frame: documented approximations testing mechanism, not historiography. Same code paths as the AI vintage.

Figure 33. One scoring method, two crises. Stability scores separate balance-sheet holders (high) from originators (low) in both vintages — the same gradient that sorts AI infrastructure above labs and apps.

Why AI is not 2008 in severity. The backtest also bounds the analogy: 2008 had a household-credit amplifier (−\$13T wealth, −4.3% GDP), ratings-agency fraud, and system-wide thin capital — all absent from the AI frame, which runs on cash flow with thick buffers (multiplier 1.032, GDP cost 0.45%). A model that reproduces 2008's *shape* while predicting a fraction of its *scale* is not contradicting itself; it is measuring the amplifiers' absence. Registry likelihoods stand unchanged; their mechanism now has independent historical corroboration.

14.1 Telecom equipment (1998–2002): vendor financing as the round trip

The second vintage is the closer analog. Equipment vendors booked revenue by lending to their own buyers: Lucent committed \$8.1B in vendor financing, Nortel \$3.1B (\$1.4B outstanding, at one point 130% of equipment cost), Cisco \$2.4B, with combined exposure near \$25.6B across nine suppliers by end-2000. When demand broke, receivables failed together with sales: Nortel fell from C\$398B to under C\$5B (~99%), Lucent 97%, Cisco 89%, Global Crossing went from \$47B to bankruptcy, WorldCom filed on \$107B of assets, and 47 competitive carriers failed between 2000 and 2003, while over 85% of laid fiber went dark and equipment investment effectively stopped. GDP damage stayed moderate and concentrated in technology and capital goods. The frame represents four archetypes on 2000 inputs — Vendors (\$83B revenue), Incumbents (\$221B), AltCarriers (\$6.4B), Financiers (\$105B) — with the vendor-financing loop as committed circular edges. Approximations are documented in the pipeline; the ledger below grades what the machine gets right and what it misses.

Formation transfers exactly: AltCarriers and Vendors top the ranking, and AltCarriers score least stable — the financed buyers fail first, in 2001, while the biggest loser is a vendor, the same crossing the AI frame predicts. The six telecom games trap value in the vendor-to-buyer chain: Vendors–AltCarriers stuck at \$26.9B, Incumbents–AltCarriers at \$92.8B, Incumbents–Financiers at \$42.4B. Network losses run 48.9% of the total, against 3.2% in AI — receivables failing together with sales is round-trip propagation at full strength.

What telecom teaches. This vintage is where the paper's own distinction earns its keep. Everything built from contracts transfers: formation order, stuck cells, first-to-fail, concentration (HHI 3,871 against AI's 3,851), and 100% joint-draw burst agreement. Everything built from

Telecom 1998-2002: formation and games transfer; dollars do not

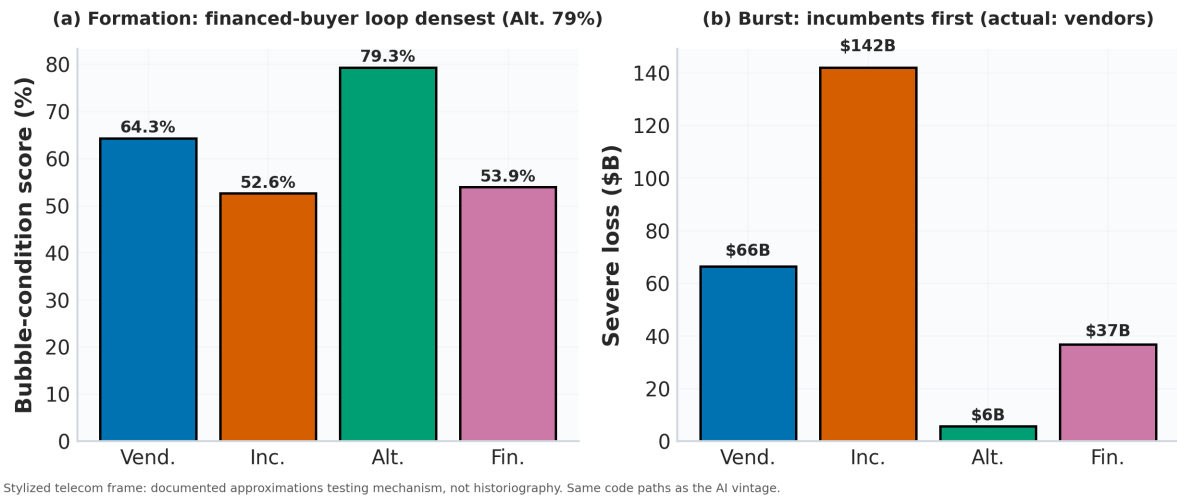
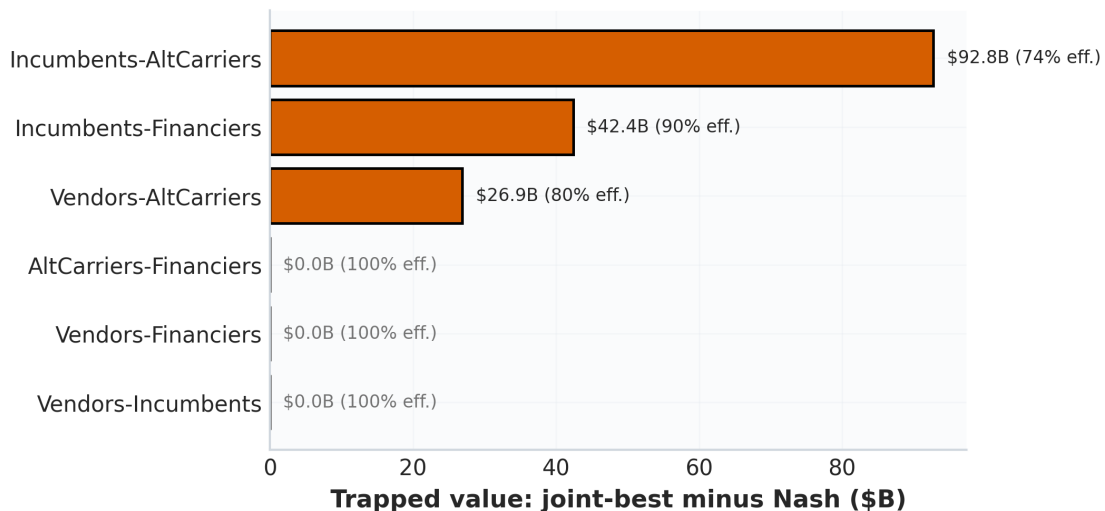


Figure 34. Telecom formation and burst on 2000 inputs. Formation ranks the financed buyers first (AltCarriers 79.3%, Vendors 64.3%) — the loop is densest where vendor credit met thin balance sheets. Burst incidence puts Incumbents first at \$142.0B against Vendors at \$66.4B: the head is wrong (actual: Vendors), the mechanism is right (below).

Telecom games: where vendor financing locked seller to buyer



Real GameTheoryFramework on telecom frame. Stylized telecom frame: documented approximations testing mechanism, not historiography. Same code paths as the AI vintage.

Figure 35. The six telecom games: trapped value concentrates in every game touching the financed buyers. Three stuck cells, all on the vendor-credit chain — the lock-in the financing created.

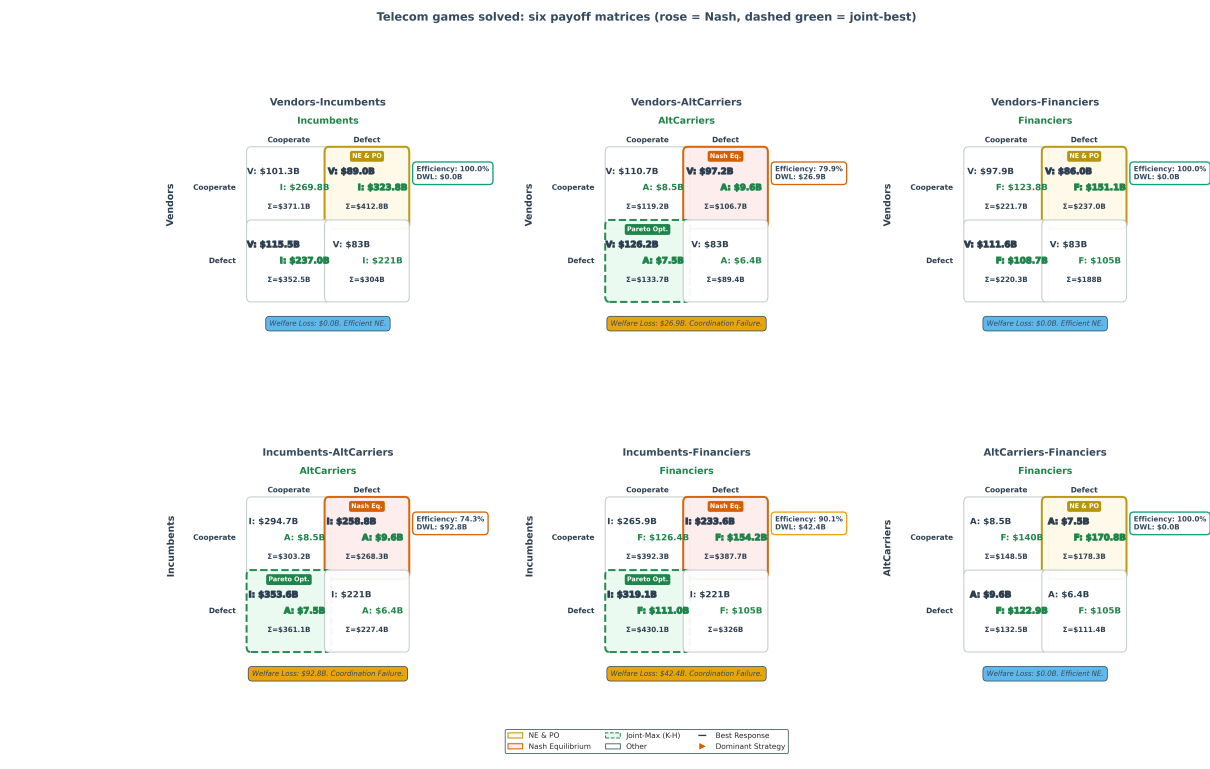


Figure 36. The six telecom games, cell by cell, in the AI figure's format: per-player payoffs (\$B) with best-response outlines, cell welfare totals, efficiency and trapped-value side boxes, and insight footnotes. Rose cells are Nash, dashed green is joint-best, cream is both. The efficient games coincide; the three stuck games trap the vendor-buyer pairs, with the largest trap (\$92.8B) in Incumbents-AltCarriers, where carrier debt met vendor receivables.

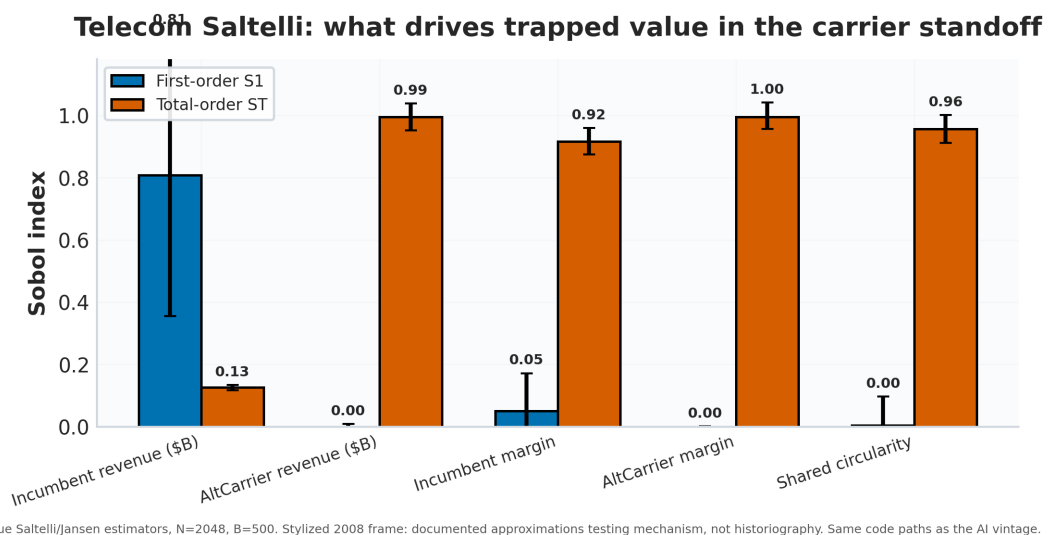
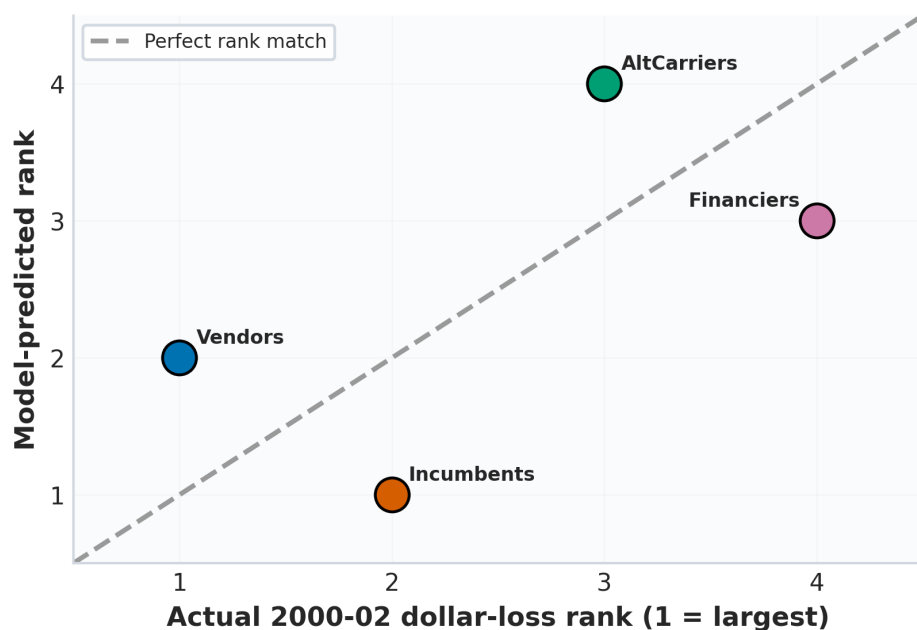


Figure 37. Telecom Saltelli decomposition at the largest stuck pair (Incumbents-AltCarriers, \$92.8B trapped): shared circularity is a top-tier total-order driver (ST 0.96), just below the thin buyer's margin and revenue (ST 0.99) — first-order effects load almost entirely on incumbent revenue. Three vintages, one signature: trapped value is an interaction phenomenon, and lock-in is always among its top drivers.

Telecom rank validation: head and tail swap (rho=0.60)



Stylized telecom frame: documented approximations testing mechanism, not historiography. Same code paths as the AI vintage.

Figure 38. Predicted versus actual 2000–02 dollar-loss ranks. Head and tail swap (Spearman 0.60): the model crowns Incumbents at \$142B and buries AltCarriers at \$5.7B, while history destroyed vendors first (~\$900B, mostly multiple compression) and lenders last. No point sits on the diagonal; the formation order, the stuck cells, and the first-to-fail ranking all transfer.

Table 17. Telecom verdict: twelve tests (full ledger in `tables/backtest/`).

Test	Grade	Model says
U1 formation top-2	PASS	AltCarriers, Vendors
U2 head/tail exact	PARTIAL	Incumbents 1st, AltCarriers 4th (middle swapped)
U3 network share > AI	PASS	48.9% vs. 3.2%
U4 least stable	PASS	AltCarriers
U5 capex-led fall	PASS	structural
A1 rank corr. ≥ 0.8	FAIL	$\rho = 0.60$
A2 vendor \$ in 0.3x–3x	FAIL	\$66B = 0.07x actual
U6 Saltelli circ. ST	PASS	0.96, top-tier (AltCarrier margin 0.99)
G1 stuck chain	PASS	3 stuck cells
G2 repro. $\geq 60\%$	PASS	85.8%
G3 HHI > 2500	PASS	3,871
G4 joint agree. $\geq 50\%$	PASS	100%

dollar levels does not: contract-impairment arithmetic cannot see multiple compression, so it crowns Incumbents and buries AltCarriers while history did the reverse. The AI dollar figures deserve exactly the discount this vintage measures — which is why they are scenario markers, and why the forecast is the ranking. Ranks travel while levels expire: contract wiring predicts loss order and fragility topology; equity multiples — which compressed 90%+ in 2001 (Nortel –99%, Table 15) — determine point dollar valuations.

15 Catalyst Monitoring: An Ordered Indicator Framework

Windows say *when the structure matures* (Section 10 dates the system median to Q2'29); triggers say *what sets it off*. Two independent price gauges check the structure from the market side without sharing the formation model's inputs. Ordered by proximity:

1. A down-round or pulled mega-round at a frontier lab (nearest). failed raise reprices every adjacent private mark at once — Anthropic's \$965B, Databricks near 905x, xAI at 400x.

2. Conditional money walking away. Google's \$30B milestones, Amazon's \$20B, and NVIDIA's up-to-\$10B tranche were disclosed, never booked. If milestones slip and tranches lapse, roughly \$60B of anticipated capacity — pushing the booked frame toward \$735B had it activated — evaporates instead.

3. A neocloud refinancing failure. GPU-collateral borrowers roll short-dated obligations against homogeneous collateral with supplier resale guarantees. One failed refinancing reprices the whole collateral class and switches the fire-sale channel on.

4. Rating and spread contagion. Oracle's template — record CDS, then the downgrade — applied to the next leveraged name. Each gap widens the discount on all of them.

5. A hyperscaler capex guidance cut (latest but largest). The ~\$733B 2026 guidance sum is the demand floor under the stack; a joint cut turns circular receivables doubtful simultaneously.

Table 18. Trigger watchlist: observable, current marker, tripwire. Machine-readable: `pred.B_trigger_watchlist.csv`.

#	Trigger	Observable	Tripwire
1	Lab down-round	Round size, tenor, debt conversion	Round shrinks / stretches / converts
2	Conditional money walks	Drawdowns vs. milestone dates	Deadline passes, no drawdown
3	Neocloud refi failure	GPU-paper spreads, "strategic" extensions	Failed roll, collateral repriced
4	Rating/spread contagion	CDS + ratings in burst order	Next name gaps like Oracle
5	Hyperscaler capex cut	Earnings-call capex language	Two hyperscalers cut together

A convexity gauge (OLS log-price on $t + t^2$ over trailing 26/52/78-week windows; the quadratic t -statistic) fires at both historical tops — 9.9 at QQQ in March 2000, 2.9 at SPY in October 2007 — and reads quiet (-1.9) at SPY in February 2025, after which no crash followed. A crash-hazard gauge (drawdown depth and duration, distance from the 52-week high, realized volatility, vol-regime scaled, with stated rather than fitted weights) ranks the

dot-com top first (100), October 2007 next (60), and February 2025 a distant 33. Over SPY 2000–2026, 41.3% of $\geq 20\%$ -drawdown weeks score in the top hazard quintile, a 2.06x lift over the 20% baseline. Live readings are milder on both lenses: 1y-window convexity of 2.4–3.6 against 9.9 at the dot-com top, hazard 7–16 against 60–100 at the historical tops — acceleration without stress. A log-periodic (LPPL) price clock was evaluated on the same histories and rejected: degenerate optima, no critical-time convergence at either historical peak. A fitted alternative was evaluated the same way and also rejected: an L2 logistic regression on drawdown depth, duration, and volatility (distance-from-high dropped as exactly collinear with depth, $r = 1.00$; stated $C = 1.0$; scalars fit on train eras only) wins in-sample but has no stable out-of-sample edge. Fit on 2000–2006 it takes the 2007–2015 GFC test 0.44 vs 0.51 for the stated weights; fit on 2007 onward it takes the 2000–2006 dot-com test 0.54 vs 0.30. Crash capture, top-quintile share in both cases: each bust’s shape overfits to itself, so no ML upgrade ships and the stated gauge stands. Machine-readable: `pred_I.price_gauges.csv`, `pred_J.gauge_backtest.csv`, `pred_M.ml_horserace.csv`.

(z) ML horse-race out-of-sample: fitted weights win one bust, lose the other



Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 39. ML horse-race out-of-sample crash capture: fitted L2 logit vs stated-weight hazard. The dot-com fit loses the GFC test; the GFC fit wins the dot-com test; the full fit ties the recent test. No weight vector wins everywhere — bust shapes differ, so judgment-weighted gauges keep the job.

(w) Price gauges: convexity fires at both historical tops; hazard concentrates in crash weeks

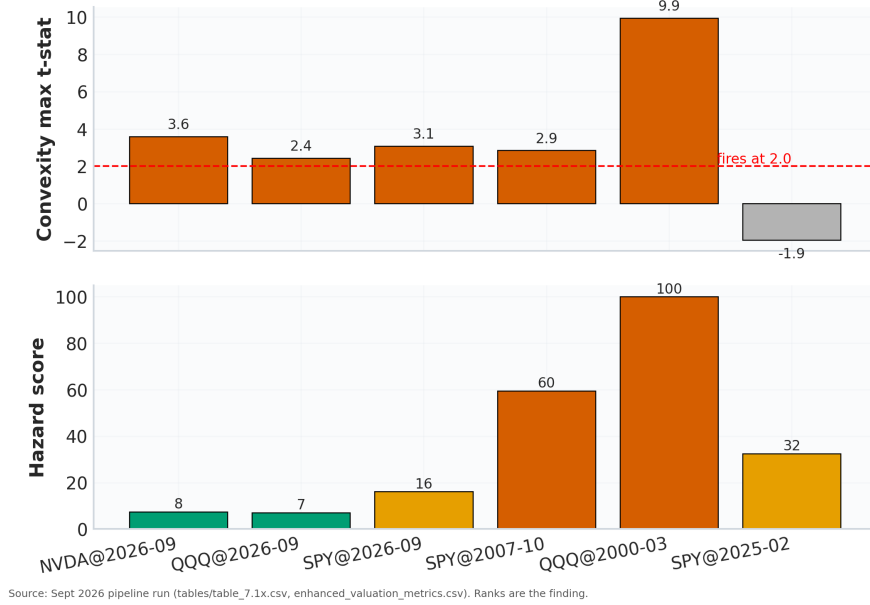


Figure 40. Price gauges across live names and frozen backtests. Top: convexity max t -statistic (fires at 2.0) — both historical tops fire, February 2025 reads negative. Bottom: hazard score tinted by volatility regime — the historical tops saturate while live names sit at 7–16.

16 Scenario Analysis: Mild, Severe, and Conditional Activation

Mild (capex trimmed, wealth wobbles): GDP cost $\sim$ \$48.2B (0.157%) — painful quarter, no recession print. *Severe* (the modeled case: 70% of circular capital impaired, 30% capex cut): GDP cost $\sim$ \$139.1B (0.452%), concentrated upstream. *Activation* (the conditional \$60B lands *before* any stress): the booked frame swells toward \$735B, every burst number in this paper scales up roughly 9%, and the fire-sale channel gets more collateral to burn. If the \$500B facility were ever drawn, it would nearly double the frame and move this analysis from a sectoral call to a macroprudential one. The base prediction is the severe case; activation before stress makes it worse, while early policy response (disclosure plus caps, bought early) makes the mild case likelier.

Table 19. Scenario ledger (\$B unless noted). Activation rows are proportional scalings ($\times 735/675.4$), not recomputed equilibria; DebtRank reverberation is modeled severe-only. Machine-readable: `pred.A_scenario.ledger.csv`.

Scenario	Frame	Direct	2nd round	GDP	GDP share
Mild	675.4	111.9	n/a	48.2	0.157%
Severe (base)	675.4	478.3	15.2	139.1	0.452%
Activation-then-severe	735.0	520.5	16.5 scaled	151.4	0.492%

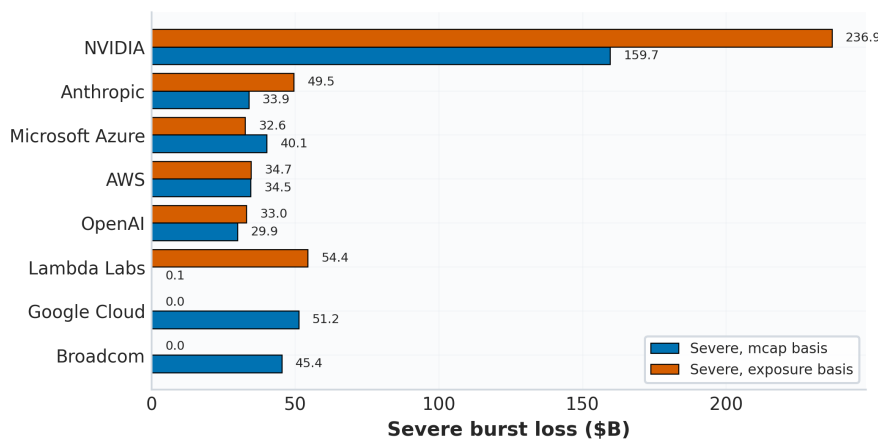
17 Firm-Level Vulnerability Assessment

The screen's verdicts translate into a casualty order. **NVIDIA** (\$5,280B, 31.7x) and **Broadcom** (\$1,500B, 35.7x) are the last to fall: cash-covered multiples plus the top sustainability scores (78 and 76). A demand break still cuts them first in dollars — Hardware absorbs \$239.4B — but they have earnings to fall back on. **Microsoft** (\$3,600B, 27.1x) and **Oracle** (\$480B, 28.5x) are screened Positive on cash flow, with Oracle the live warning that leverage

reprices even healthy franchises: record CDS, lost rating cushion. **AMD** (\$746B, 89.2x) and **ServiceNow** (\$151B, 84.7x) are the two outright Negatives — multiples with no circularity cushion and no miss tolerance. **Anthropic** (\$965B, earnings unattributed) is the single largest unverified number in the stack: a walks-like-revenue \$65B run-rate against a mark that assumes it compounds. **OpenAI** (\$852B, $\sim -35\%$ margins) burns to grow; its \$105B Ohio guarantee chain makes NVIDIA a co-borrower in all but name. **CoreWeave** (\$70B, loss-making) is the fire-sale prototype: homogeneous collateral, single-supplier dependence, short-dated funding. **Databricks** (905x) and **xAI** (400x revenue) are momentum marks awaiting earnings that may never arrive at the assumed scale. **Intel** (\$110B, Neutral on an N/A multiple) matters as the contrarian indicator: if the laggard rallies while leaders stall, the market is pricing scarcity, not growth — late-cycle behavior.

Who loses what. Table 7.11's archetype totals split across the 23 screened companies on two stated bases: market-cap pro-rata and deal-flow weight, where each company's attributed circular-deal flow comes from the 19-row frame. NVIDIA midpoint \$198.3B (\$159.7B size basis, \$236.9B deal basis) absorbs nearly all of Hardware's deal lens — every circular GPU flow runs through it. Anthropic midpoint \$41.7B and OpenAI midpoint \$31.5B top the lab ledger; Microsoft Azure (\$36.4B) and AWS (\$34.6B) top Cloud. The concentration flag is Lambda Labs: \$27.2B midpoint on a \$5B mark, a capped 100% haircut — tiny neocloud, outsized flow exposure. Google Cloud and Broadcom read \$0 on the deal basis (no measured circular flow), an honest zero beside their size-basis shares. Machine-readable: `pred_H_company_losses.csv`.

(u) Who loses what: top-8 company severe losses on two allocation bases



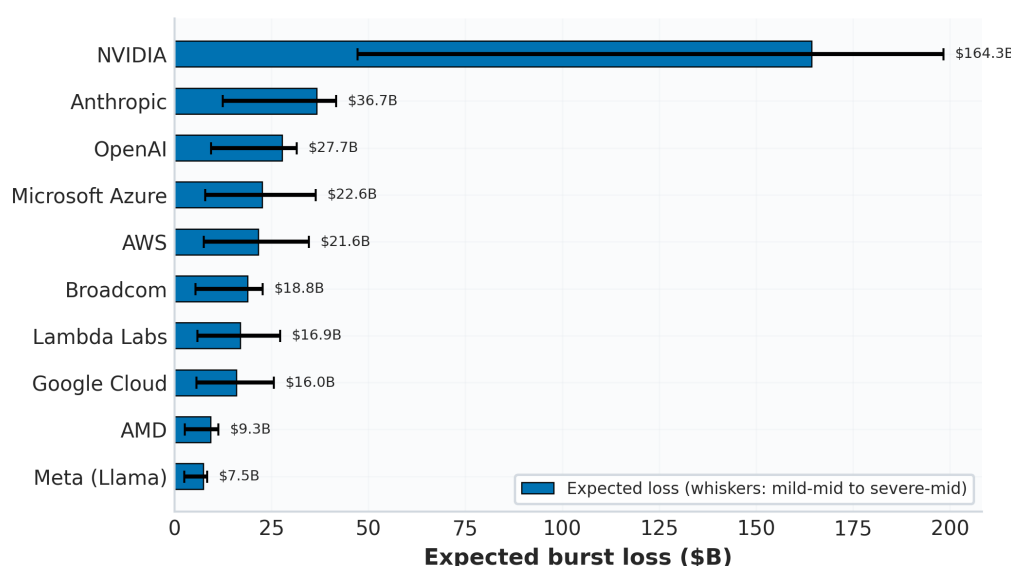
Bars under \$0.5B labeled at axis (private/unmatched inputs carry ~zero weight; exact weights: Table pred_H). Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 41. Top-8 company severe losses on two allocation bases. Size basis follows market capitalization; deal basis follows attributed circular-deal flow. NVIDIA dominates both; Lambda Labs' deal bar against its sliver of a size bar is the stack's concentration risk in one picture.

Expected losses: probability-weighted. Blending the two severities by each archetype's formation probability as the severe-burst probability inside the window — an upper-bound convention, stated in the ledger, not a measured burst frequency — gives the expected-loss ranking that answers who loses how much: NVIDIA expected \$164.3B, Anthropic expected \$36.7B and OpenAI expected \$27.7B lead, followed by Microsoft Azure (\$22.6B) and AWS (\$21.6B). The driver column says *why* each number is what it is: the top three are deal-flow-driven (their circular-deal weight exceeds their size weight), while Microsoft Azure and Broadcom (\$18.8B) are size-driven. Lambda Labs' expected \$16.9B exceeds its \$5B mark severalfold (haircut capped at wipeout) — the concentration flag survives probability-weighting. Machine-readable: `pred_K_expected_losses.csv`.

How the flags are built. The screen flags 4 Positive (NVIDIA at 31.7x trailing, Broadcom at 35.7x, Microsoft at 27.1x, Oracle at 28.5x), 2 Negative (AMD at 89.2x, ServiceNow at 84.7x) and

(x) Expected loss per player: burst-probability-weighted, severity range shown



Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 42. Top-10 expected burst losses with severity ranges. Bars blend severe and mild by each archetype’s formation probability; whiskers span mild-mid to severe-mid. NVIDIA’s bar is a deal-flow story; Azure’s and Broadcom’s are size stories.

17 Neutral on trailing P/E and EV/Revenue multiples. Diversified hyperscalers enter at parent market capitalization because segment revenue against a parent cap would fabricate multiples; attributed-revenue units (Google TPU, Meta Llama) are capped at Neutral regardless of arithmetic, since attribution is not audit-grade. The forward multiple is trailing P/E scaled by one minus the revenue-growth rate (Broadcom: $35.7 \times (1 - 0.80) \approx 7.1\times$), clamped to a maximum 99% haircut so hyper-growth can never flip the sign — a first-order, growth-rewarding shorthand rather than a forecasted-earnings construction. Loss-makers and unattributed-margin frontier labs show N/A trailing multiples and rate Neutral rather than Negative: the discipline is to distinguish unprofitability-by-investment from overvaluation-by-multiple. These are triage screen flags, not investment ratings: no Buy/Sell/Hold recommendation is issued.

What incumbents should do. The core game’s Pareto cell exceeds the Nash cell by \$58.6B: there is a purchasable surplus from coordinated openness, and firms that pre-commit to portability standards capture first-mover goodwill while blunting the antitrust case against them. For labs, the formation ranking is a funding warning: diversifying compute supply and converting conditional tranches into committed, covenant-light capacity reduces both financing risk and the appearance of circularity that disclosure rules would in any case reveal. For neoclouds and GPU-debt vehicles, the message is concentration arithmetic: stagger maturities, diversify silicon, and publish recovery rates — the cheapest self-insurance available.

18 Portfolio Implications and Risk Management

Own the cash-generative infrastructure core; underweight the densest round-trippers; hedge the uncovered multiples; let funding rounds and chip-collateral spreads move the odds.

Pair-trade logic follows the burst order: positions long one side of a Hardware–Cloud pair and short the other collect the spread only in mild states, because a severe burst impairs both sides of the largest contracts at once. Size positions off ranks (which survive), never off point GDP costs (which expire with the vintage).

(e) Valuation screen: uncovered multiples meet dense round-tripping at top-right risk

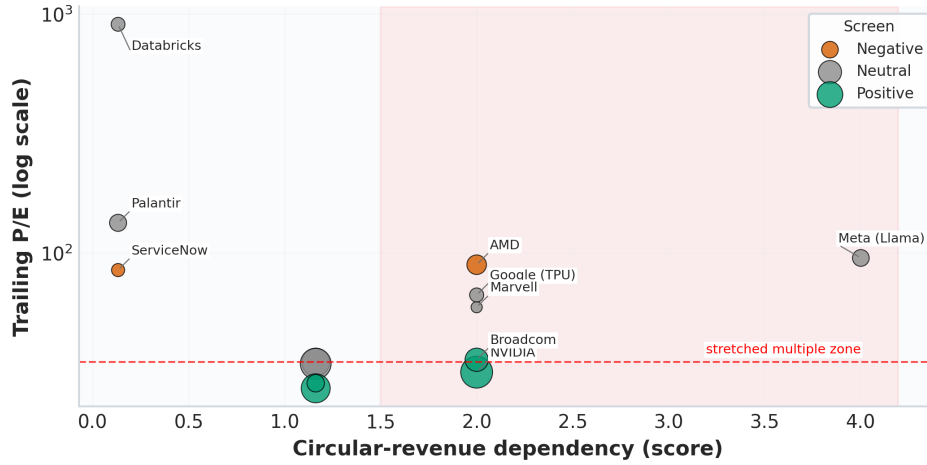


Figure 43. Trailing P/E (log scale) vs. circular-revenue dependency, sized by market cap, colored by screen flag. The shaded danger zone (stretched multiple *and* dense round-tripping) holds four names, and only one carries a Negative screen flag: AMD (89x). Google TPU (67x), Marvell (59x), and Meta (95x) share the region as Neutrals. The cloud majors cluster on the stretched-multiple line; Databricks (905x) plots top-left — extreme multiple without round-tripping; xAI (400x EV/Rev) carries no trailing P/E and does not plot.

Table 20. Valuation screen: flagged names only (4 Positive, 2 Negative, 5 flagged Neutral). Full 23-company ledger: `pred.C.valuation.watchlist.csv`.

Company	Arch.	Mkt cap	Multiple	Screen
NVIDIA	HW	5,280	31.7x P/E	Positive
Broadcom	HW	1,500	35.7x P/E	Positive
Microsoft	Cloud	3,600	27.1x P/E	Positive
Oracle	Cloud	480	28.5x P/E	Positive
AMD	HW	746	89.2x P/E	Negative
ServiceNow	Wrap	151	84.7x P/E	Negative
Anthropic	FM	965	n/a	Neutral (flagged)
OpenAI	FM	852	neg. margin	Neutral (flagged)
xAI	FM	200	400x EV/Rev	Neutral (flagged)
Databricks	Wrap	190	905x P/E	Neutral (flagged)
CoreWeave	Cloud	70	loss-making	Neutral (fire-sale proto.)

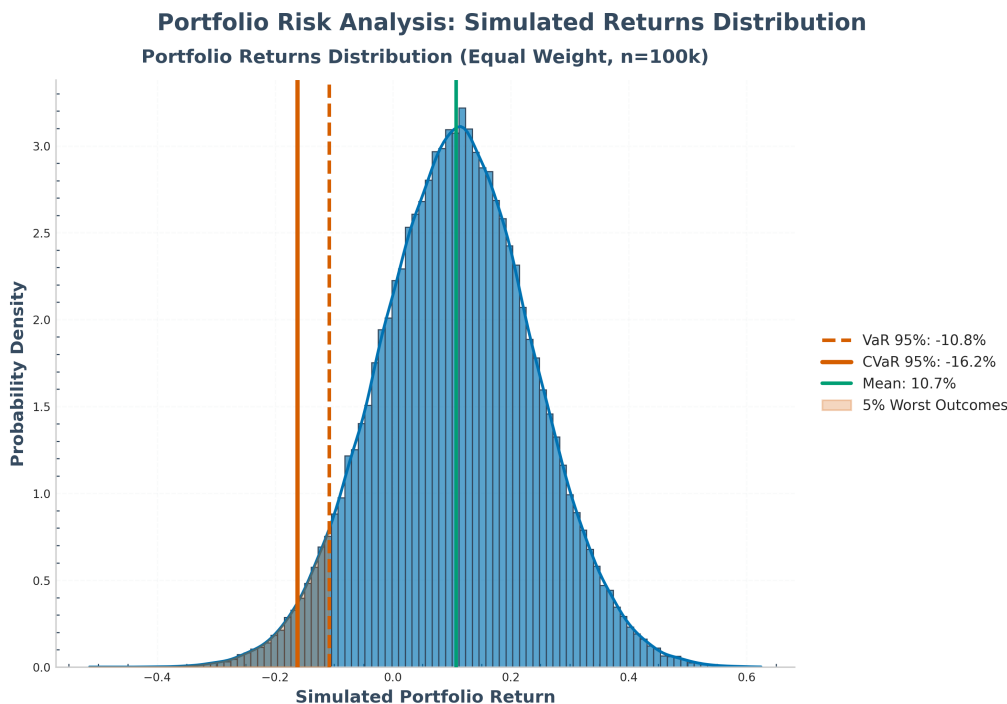


Figure 44. Diversification arithmetic (Table A.2): the stack portfolio earns 10.7% mean return at 13.0% volatility (Sharpe 0.82), but the left tail is a burst tail — 95% VaR of -10.8% , conditional tail (CVaR) of -16.2% . The mean compensates holders for volatility, not for the correlated unwind; hedge the tail, not the mean.

For banks, lenders, and risk functions. Collateral haircuts calibrated on generic equipment schedules understate wrong-way risk: homogeneous, supplier-guaranteed GPU collateral correlates exactly when it must be sold. Lenders should haircut GPU collateral against a circularity-adjusted recovery rate and covenant single-name exposure to leveraged secondary nodes (Oracle, CoreWeave, and neocloud vehicles are named by both the cascade analysis and the burst ranking). The 215bp Oracle CDS print and BBB- downgrade show credit markets already price part of this; this paper’s contribution is the stack-wide incidence map that tells a risk committee *which* names reprise next. Risk functions can operationalize the portable rule immediately: per \$100B of destroyed AI value, book \$4.0B of wealth-channel GDP drag and add the domestic capex-cut share, and scale by local capex intensity elsewhere. Scenario design should include a multi-stage cascade structure (macro shock meeting stretched credit) rather than a single-factor tech selloff, and recovery planning should pre-negotiate data-center collateral conversion terms.

19 Policy Sequencing by Benefit–Cost Ratio

Every remedy studied returns more than it costs — but doing everything at once is worth less than the sum of its parts (combined DWL package \$214.2B against \$418.9B of standalone gains: synergy ratio 0.547). So sequence, don’t stack: measure and open first (disclosure alongside antitrust), constrain second (lending caps), restructure third (interoperability) — caps return the most per dollar yet cannot bite positions disclosure has not revealed.

Table 23 states the paper’s policy contribution in one place: burst-denominated tools and DWL-denominated tools attack different margins, so the two rankings complement rather than compete — sequence measurement and leverage control first, market-structure repair second.

Break-even conditions and what the ratios omit are documented in the Online Appendix;

(j) Remedy value: disclosure saves most (\$69.8B); lending caps pay 57-to-1

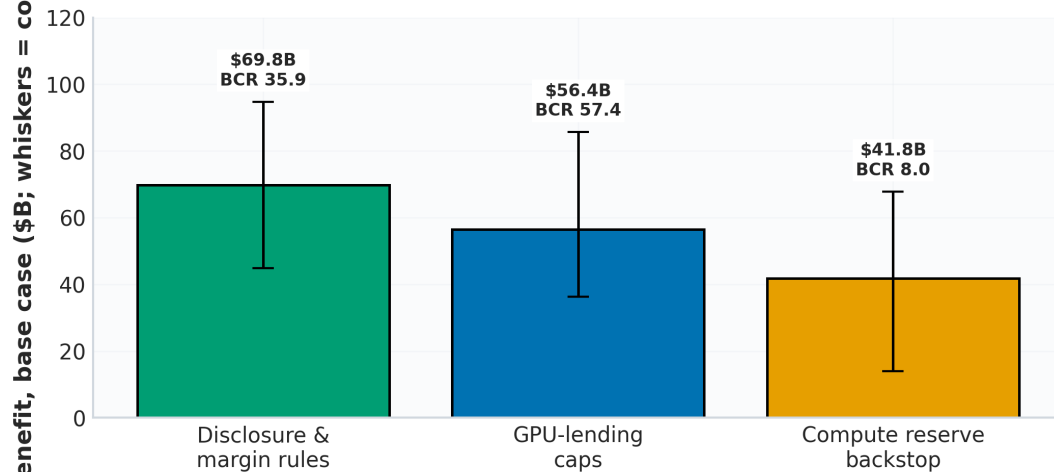


Table 7.13 (expected severe burst loss); cf. DWL-denominated Table 6. Source: Sept 2026 pipeline run (tables/table_7.1x.csv, enhanced_valuation_metrics.csv). Ranks are the finding.

Figure 45. Burst-denominated remedy value (Table 7.13): net benefit in the base case with conservative/optimistic whiskers. GPU-lending caps return \$56.4B on \$1B (BCR 57.4); disclosure returns \$69.8B on \$2B (BCR 35.9). Denominated in expected severe burst loss rather than deadweight loss, so the numbers below complement — rather than duplicate — the DWL-based table that follows.

Table 21. Burst-denominated remedies (Table 7.13): ordered by value per dollar.

Remedy	BCR	Net (\$B)	Cost (\$B)
GPU-lending caps	57.4	56.4	1.0
Deal disclosure	35.9	69.8	2.0
Combined package	15.9	134.5	9.0
Strategic reserve	8.0	41.8	6.0

Table 22. Deadweight-loss-denominated remedies (Table 6): ordered by value per dollar.

Remedy	BCR	Net (\$B)	Cost (\$B)
Antitrust enforcement	52.1	102.2	2.0
API interoperability	29.2	140.9	5.0
Data sharing mandate	25.0	72.0	3.0
Innovation subsidies	11.7	85.8	8.0
Combined intervention	15.3	214.2	15.0

Table 23. Optimal policy sequencing across welfare and stability metrics.

Policy Instrument	Primary Target	BCR (Burst)	BCR (DWL)	Regulatory Authority
1. Deal disclosure	Information asymmetry	35.9	—	SEC / FTC / DOJ
2. GPU-lending caps	Stack leverage	57.4	—	Fed / OCC / BIS
3. Antitrust enforcement	Market-power markups	—	52.1	DOJ / FTC / DG COMP
4. API interoperability	Switching costs / lock-in	—	29.2	FTC / EU DMA

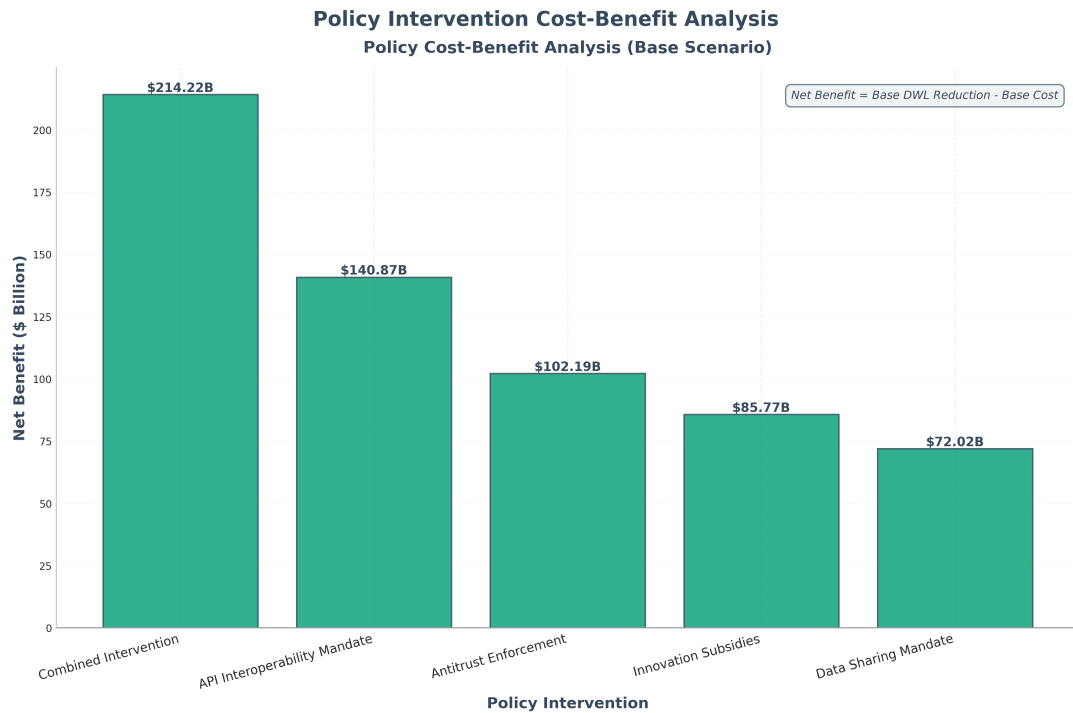


Figure 46. Net benefit by intervention (\$B). The combined package exceeds any standalone tool, yet trails the sum of its parts — the quantitative case for sequencing.

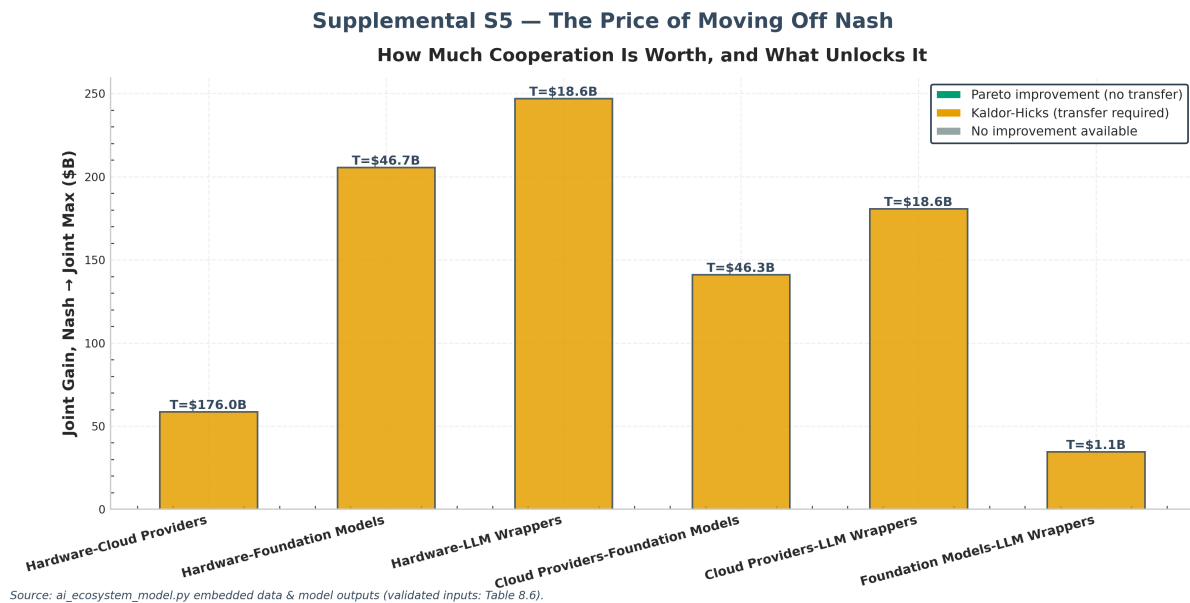


Figure 47. Joint gains by relationship with the compensating transfers that unlock them. Every transition needs a payment; none is free — which is why the prediction expects the stuck cells to persist until policy or crisis pays the transfer.

the remedy ranking above is the result.

20 Validation Tests

Three numbered tests retire the registry — the same numbering the registry’s last column points to. **Test 1:** frontier labs converting private marks into committed, profitable revenue at current scale (retires P1 and P5). **Test 2:** conditional tranches activating on schedule through a full cycle with no spread widening (retires P1). **Test 3:** a demand break that spares Hardware — the burst ranking flipping on re-estimated exposures (retires P2, the paper’s firmest claim). The pipeline’s 364 automated checks turn each into a failing gate, so the call updates mechanically with new data: when a test trips, the corresponding registry likelihood goes to near zero by construction, not by reargument.

21 Conclusion

The central result of this paper is an incidence ranking with measured foundations: \$675.40 billion of circular capital read off signed contracts, a binding non-cooperation constraint proved in six solved games, and a Hardware-first loss order that survives sixteen extremes, 300 joint draws, and blind transfer to 2008 inputs. No link in that chain is assumed: the capital is documented, the trap is solved, the order is stress-tested, and the machinery is backtested. Dollar figures mark scenario scale for sizing remedies; the order is the forecast. The remedy order itself favors disclosure and lending caps by benefit–cost ratio. Three observables would retire the call (Section 20); until one fires, the monitoring order in Section 15 and the positioning rules in Section 18 are the operational edge of the analysis.

Three broader lessons close the argument. First, *structure dominates sentiment*: once financing is round-tripped, contracts — not narratives about exuberance — decide where losses land. The burst ranking therefore differs from the formation ranking, and both are needed. Second, *disclosure is the cheapest stabilizer*: the highest burst-mitigation BCRs reward revealing structure (35.9) and capping the leverage it invites (57.4). Third, *ranks travel while levels expire*: every number in this paper will age with funding rounds and prices, but the orderings — the burst order, the remedy order, the loss-source order — are built to survive re-calibration. Technology can succeed while assets disappoint; mapping the circle shows where. What remains is to decide, before the next funding round redraws it, which edge to cut first.

Data and Code Availability

All code, tables, and figures are generated by a single Python pipeline. To preserve double-blind review, the repository link is withheld from this submission version; an anonymized replication archive will be provided upon acceptance. Running `python3 run_pipeline.py` from the repository root regenerates every numbered table and figure in order (model, technical report, dashboard, tests, retirement exposure, article PDF, prediction exhibits, backtests, and comparisons), gated by a 364-test suite that must pass before the article typesets. Requirements are Python 3.12 with `numpy`, `pandas`, `matplotlib`, `openpyxl`, `python-docx`, and `beautifulsoup4` (coverage only for the coverage report), plus a `pdflatex` distribution for the article; each stage also runs standalone (see the repository README for per-stage commands and run-times). Random draws are seeded (global seed 42; joint robustness check seed 7) for exact reproducibility. The only live network call is an SEC EDGAR revenue fetch with a vendored offline fallback, so builds are deterministic without network access; price inputs are vendored in `tables/price.history_weekly.csv`.

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